Forecasting Demand for National Research Supercomputing

Australian Academy of Science | Australian HPC Demand to 2030

Yusuf Kurnia Romadhon

Table of contents

1 Introduction and Objectives **1**
1.1 Project Background and Forecasting Problem . . . 1
1.2 Project Objectives . . . 1

2 Data Sources and Preparation **2**
2.1 Data Sources and Coverage . . . 2
2.2 Data Integration, Standardisation and Provenance . . . 3
2.3 Variable Definitions and Measurement Units . . . 3
2.4 Compute Resource Conversion . . . 4
2.5 Field of Research Classification Alignment . . . 5

3 Forecasting Methodology **6**
3.1 Forecasting Framework . . . 6
3.2 Forecasting Models . . . 7
3.3 Forecasting Targets and Historical Coverage . . . 8
3.4 Model Evaluation . . . 8
3.5 Model Selection and Sensitivity Analysis . . . 9
3.6 Field of Research Scenario Development . . . 10
3.7 Forecast Horizon and Interpretation . . . 11

4 Results **11**
4.1 Initial Analysis . . . 11
4.2 Forecast Model Evaluation . . . 16
4.3 Demand Forecasts to 2030 . . . 23
4.4 Field of Research Scenarios . . . 31
4.5 Summary of Results . . . 35

5 Discussion **36**
5.1 Interpretation of the Main Findings . . . 36
5.2 Implications for National HPC Planning . . . 38
5.3 Limitations and Future Work . . . 39

6 Conclusion **41**

7 References **41**

1 Introduction and Objectives

1.1 Project Background and Forecasting Problem

High-performance computing (HPC) supports research that requires substantial computing resources, including scientific simulations and large-scale data analysis (Australian Academy of Science 2024). In Australia, the National Computational Infrastructure (NCI) and the Pawsey Supercomputing Research Centre operate the two national Tier-1 HPC facilities available to researchers (National Computational Infrastructure, n.d.-b; Pawsey Supercomputing Research Centre, n.d.-a).

This project was undertaken with the Australian Academy of Science to examine historical research computing demand and forecast national HPC demand through 2030. It uses publicly available records from NCI, Pawsey and the National Computational Merit Allocation Scheme (NCMAS) to analyse requested and allocated computing resources. The results are intended to inform discussions about future national computing infrastructure and public investment (Australian Academy of Science 2026).

Forecasting demand is difficult because allocated computing resources do not necessarily represent what researchers need. Allocation schemes operate within available capacity, so researchers may receive less computing time than they request (National Computational Infrastructure 2014b, 2020b). Requested compute also has limitations, as researchers may restrict their applications based on expected availability or choose not to apply. The analysis therefore distinguishes requested compute, allocated compute and the gap between them, while recognising that underlying demand cannot be measured directly from these records (Australian Academy of Science 2026).

Historical comparisons are further affected by changes in computing systems, reporting practices and service-unit accounting. CPU and GPU resources are measured differently, while revisions to Field of Research (FoR) classifications affect comparisons across years (Australian Bureau of Statistics 2020). Given the limited annual observations, the analysis distinguishes directly reported values from estimates and converted measures when preparing the historical dataset and interpreting forecasts.

1.2 Project Objectives

The main objective is to forecast Australian national research computing demand through 2030 using available historical records, while documenting the limitations of the data and the assumptions required for comparison.

The project has five objectives:

1. **Construct a historical dataset** of HPC demand and allocation from publicly available NCI, Pawsey and NCMAS records, with documented sources and reporting limitations.
2. **Standardise historical measures** by examining changes in computing systems, resource units and reporting practices, and applying conversion methods where necessary.
3. **Analyse demand and allocation patterns**, including requested and allocated resources, the gap between them, CPU and GPU demand, and FoR allocation composition.
4. **Develop and evaluate forecasting models** using the available annual observations, comparing forecast accuracy, model assumptions and sensitivity to recent data.

5. **Develop demand scenarios through 2030** and examine how different growth assumptions and historical FoR allocation patterns affect the results.

These objectives are consistent with the data analysis, forecasting and scenario development requirements in the internship brief (Australian Academy of Science 2026).

2 Data Sources and Preparation

2.1 Data Sources and Coverage

The analysis covers 13 financial years, from 2013/14 to 2025/26, represented by analytical years 2014–2026. Earlier data were assembled primarily from National Computational Infrastructure (NCI) annual reports and Pawsey Supercomputing Research Centre publications (National Computational Infrastructure 2014a, 2015, 2016, 2017, 2018, 2019, 2020a; Pawsey Supercomputing Research Centre 2017). More recent data were drawn from National Computational Merit Allocation Scheme (NCMAS) outcomes (National Computational Merit Allocation Scheme 2021, 2022, 2023, 2024, 2025, 2026).

The sources provide different levels of detail. Earlier publications mainly contain annual or facility-level summaries, while more recent NCMAS outcomes include project-level allocations, computing resources and research fields (National Computational Merit Allocation Scheme 2021, 2024, 2026). Different documents sometimes supplied requested compute, allocations and FoR information for the same year. Historical applicant guidance was also reviewed to understand changes in allocation rules and resource definitions (National Computational Infrastructure 2014b, 2020b, 2022, 2023, 2025).

Table 1: Main data sources and their analytical roles.

Source	Information extracted	Use in analysis
NCI annual reports	Historical allocations, computing systems and reported resources	Historical series construction
Pawsey publications	Allocation information, computing systems and resource usage	Supplementary facility evidence and cross-checks
NCMAS outcomes	Requested and allocated resources, projects and research fields	Demand, allocation and FoR analysis
Supporting publications	Accounting rules, resource descriptions and contextual indicators	Conversion assumptions and interpretation

Research funding, workforce and research activity indicators were considered as potential predictors but were not included in the final models because sufficiently consistent annual series were unavailable.

2.2 Data Integration, Standardisation and Provenance

Information from NCI, Pawsey and NCMAS publications was organised into two datasets: an annual demand and allocation summary, and a Field of Research (FoR) allocation composition dataset. Records

were aligned by reporting year, computing system and resource type, with detailed observations checked against published annual totals where available.

Financial years were assigned to their ending year; for example, 2013/14 was recorded as 2014. Computing system names, resource identifiers and research field labels were standardised to support comparisons across reporting periods. CPU, GPU and aggregate computing resources were identified separately, and differences in reporting scope were recorded before aggregation.

Original values, measurement units and source references were retained alongside standardised fields. Each observation was identified as directly reported, calculated from source information or estimated using documented assumptions. Missing resource breakdowns were not treated as observed values.

2.3 Variable Definitions and Measurement Units

The main annual variables are requested compute and allocated compute. Where both measures are available on a comparable basis, the request-to-allocation ratio is calculated as:

$$R_t = \frac{D_t}{A_t}$$

where D_t represents requested compute and A_t represents allocated compute in year t.

The corresponding fulfilment rate is A_t/D_t, and the request–allocation gap is $D_t - A_t$. All three indicators are calculated only when the values have compatible units and reporting scope (Australian Academy of Science 2026).

Historical records use different measurement units, including service units (SU), kilo service units (kSU), million compute hours, CPU core-hours and GPU hours. Service units are accounting measures that depend on the computing system and resource type. Consequently, values expressed in kSU are not necessarily computationally equivalent across facilities.

CPU core-hours and GPU hours were treated as separate measures.

2.4 Compute Resource Conversion

Historical service-unit values were converted using accounting factors specific to each computing system and resource type (National Computational Infrastructure, n.d.-b; Pawsey Supercomputing Research Centre, n.d.-a). Original values were retained, and conversions based on assumptions were identified separately from directly reported measures.

2.4.1 Conversion Formulas

The general relationship between service units and kilo service units is:

$$\text{SU} = \text{kSU} \times 1{,}000$$

For CPU resources reported in kSU, CPU core-hours were calculated as:

$$\text{CPU core-hours} = \frac{\text{CPU kSU} \times 1{,}000}{\text{SU per CPU core-hour}}$$

For GPU resources reported in kSU, GPU hours were calculated as:

$$\text{GPU hours} = \frac{\text{GPU kSU} \times 1{,}000}{\text{SU per GPU hour}}$$

The denominators represent the accounting factors applicable to each computing system and resource type. These factors were not assumed to be constant across all systems or reporting periods.

Where values were already reported in CPU core-hours or GPU hours, they were retained without additional service-unit conversion. Values reported in millions of hours were multiplied by 1,000,000, subject to confirmation of the original resource definition.

2.4.2 System-Specific Conversion Rules

Table 2: Main compute-resource conversion rules recorded in the year-by-year conversion worksheet.

System or resource	Reporting period	SU per resource-hour	Treatment
Raijin CPU and Magnus CPU	Selected historical years from 2013/14	1	Historical baseline; assumptions apply to some records
Gadi CPU	2020/21–2025/26	2	System-specific accounting factor
Setonix CPU	2021/22–2025/26	1	System-specific accounting factor
Setonix GPU	2021/22–2025/26	128	GPU accounting factor
Setonix Q	2025/26	256	GH200 GPU-hour conversion
Previously reported million core-hours	Selected historical years	Not applicable	Multiply by 1,000,000

The factors were compiled from system-specific documentation and the project's recorded conversion assumptions (National Computational Infrastructure, n.d.-a, 2025; Pawsey Supercomputing Research Centre, n.d.-c, n.d.-b). They were applied to the corresponding system and reporting period, rather than indiscriminately to aggregate kSU values without a resource breakdown.

For the earlier years, a factor of 1 SU per CPU core-hour was used as a baseline assumption where a more specific accounting factor was unavailable. For some 2019/20 records, the original technical conversion used a documented working assumption of 100% Gadi CPU-equivalent at 2 SU per CPU core-hour because a reliable CPU/GPU split was unavailable. This assumption should not be interpreted as a directly observed resource breakdown.

Historical Fornax GPU records could not be converted reliably using the available information. These observations were retained with their original units rather than assigned an unsupported conversion factor.

2.4.3 Conversion Validation and Limitations

Converted values were checked against original units, system-level records and published annual totals where comparable data were available. Aggregate service-unit values were converted into CPU core-hours or GPU hours only when resource breakdowns and accounting factors were available, or when a documented assumption was used.

Missing GPU observations were not treated as zero demand. Conversion assumptions and estimated values were retained in the technical documentation, together with the year-by-year conversion worksheet and validation code.

2.5 Field of Research Classification Alignment

The FoR data were standardised in two stages to support comparisons across 2014–2026.

The first stage addressed differences between the FoR 2008-based reporting period (2014–2020) and the FoR 2020-based period (2021–2026) (Australian Bureau of Statistics 2020). Selected historical fields were combined into broader reporting categories through `for_analysis_group`, while the existing standardised categories were retained for the later period.

The second stage established nine analytical groups for longitudinal comparison through `for_visual_group`. These groups were informed by the broader research-field categories used in NCI and Pawsey annual reporting. Related categories were combined where reporting detail differed across years, rather than assuming that the two FoR classification systems were directly equivalent.

Original field labels were cleaned by removing numerical prefixes, standardising case, replacing ampersands and removing excess whitespace. The cleaned labels were then assigned to the nine groups using explicit matching rules.

Table 3 summarises the analytical grouping applied to the reported research fields.

Table 3: Analytical FoR grouping rules used for longitudinal comparison across 2014–2026.

Analytical group	Reported fields included
Physical Sciences	Physical sciences and radioastronomy
Chemical Sciences	Chemical sciences
Earth & Environmental Sciences	Earth sciences, environmental sciences and geosciences
Engineering & Technology	Engineering and technology
Biological & Agricultural Sciences	Biological, agricultural, veterinary and food sciences
Medical & Health Sciences	Medical, health, biomedical and clinical sciences
Mathematics & Computing	Mathematical sciences, computing and information sciences

Analytical group	Reported fields included
Social Sciences, Humanities & Built Environment	Psychology, economics, humanities, social sciences, design and built environment
Unclassified & Operational	Other, unreported, unclassified and Pawsey operational categories

The groupings were developed for analytical consistency and do not represent official one-to-one conversions between ANZSRC 2008 and ANZSRC 2020. Original research field labels and allocation values were preserved.

The 2016/17 Pawsey records contain facility-wide usage information, which differs in scope from the NCMAS allocation data used for other years (Pawsey Supercomputing Research Centre 2017). This difference was retained when preparing the historical FoR dataset and is considered when comparing allocation shares across years.

Validation checks confirmed that all 133 FoR records had the required classification and allocation information, all reported fields were assigned to one of the nine analytical groups, and annual allocation shares summed to 100%.

The complete mapping rules and validation code are available in the technical Quarto documentation.

3 Forecasting Methodology

3.1 Forecasting Framework

Overall HPC demand (kSU), CPU demand (core-hours) and GPU demand (GPU-hours) were forecast separately because their measurement conventions and historical coverage differ. Field of Research (FoR) scenarios were developed from historical allocation shares because consistent field-level demand records were unavailable.

Given the limited annual observations, the analysis used naïve, drift, linear trend and log-linear trend models to compare different growth assumptions. Model selection considered forecast accuracy, interpretability and suitability for projections through 2030 (Hyndman and Athanasopoulos 2021).

3.2 Forecasting Models

Four forecasting approaches were considered: naïve forecasting, drift, linear trend and log-linear trend. These are established approaches to time series forecasting and regression-based trend modelling (Hyndman and Athanasopoulos 2021). The naïve and drift methods provided benchmarks against which the trend models could be evaluated.

3.2.1 Naïve and Drift Benchmarks

The naïve method assumes that the next observation will equal the latest observed value (Hyndman and Athanasopoulos 2021):

$$\hat{y}_{T+h} = y_T$$

where y_T is the most recent observation and h is the forecast horizon.

The drift method extends the average annual change observed between the first and last training observations (Hyndman and Athanasopoulos 2021):

$$\hat{y}_{T+h} = y_T + h\left(\frac{y_T - y_1}{T-1}\right)$$

These benchmarks provided a comparison with methods that explicitly estimate a trend from the full training sample.

3.2.2 Linear Trend Model

The linear trend model assumes that demand changes by a constant absolute amount each year (Hyndman and Athanasopoulos 2021):

$$y_t = \beta_0 + \beta_1 t + \varepsilon_t$$

where y_t represents demand in year t, β_0 is the intercept, β_1 is the estimated annual change and ε_t is the error term.

This specification is straightforward to interpret. For example, when CPU demand is expressed in millions of core-hours, the estimated trend coefficient represents the average annual increase in that unit.

However, a linear trend assumes that the historical absolute change continues over the forecast period. It does not account directly for future changes in computing technology, allocation policies or research funding.

3.2.3 Log-Linear Trend Model

The log-linear model assumes approximately constant proportional growth (Hyndman and Athanasopoulos 2021):

$$\log(y_t) = \beta_0 + \beta_1 t + \varepsilon_t$$

The corresponding growth rate is:

$$g = (e^{\beta_1} - 1) \times 100\%$$

where g is the implied annual percentage growth rate.

Unlike the linear model, the log-linear specification allows the absolute annual increase to grow as demand rises. This can produce substantially higher long-term projections when historical growth is strong.

The model requires positive demand observations and assumes that proportional growth remains relevant over the forecast horizon. Its projections were therefore considered alongside the linear model rather than accepted solely on the basis of historical fit.

3.3 Forecasting Targets and Historical Coverage

The forecasting models were applied separately to the available demand measures.

Table 4: Forecasting targets and historical coverage.

Forecasting target	Historical period	Measurement unit	Main consideration
Overall HPC demand	2014–2026	kSU	Historical reporting and accounting comparability
CPU demand	2014–2026	Million CPU core-hours	System-specific conversion assumptions
GPU demand	2022–2026	Million GPU-hours	Shorter series and estimated resource-level values

The HPC and CPU series contain 13 annual observations covering 2014–2026. GPU modelling was restricted to five observations from 2022–2026 because earlier GPU-hour estimates relied on less-supported conversion assumptions. The recent GPU series also includes derived values, limiting the reliability of model evaluation and forecasts.

All models were fitted separately for each demand measure. Forecasts were expressed in the corresponding units rather than combining CPU core-hours and GPU hours.

3.4 Model Evaluation

3.4.1 Rolling-Origin Cross-Validation

Forecasting performance was assessed using expanding-window, one-step-ahead rolling-origin cross-validation. This approach evaluates forecasts against observations that were not available when the models were fitted (Tashman 2000; Hyndman and Athanasopoulos 2021).

Under this approach, a model was fitted using the observations available up to a particular year and then used to predict the following year. The training window was expanded by one observation for each subsequent evaluation.

This approach preserves the chronological order of the data and avoids using future observations to estimate earlier forecasts.

For overall HPC and CPU demand, the initial training period covered 2014–2020. One-step-ahead forecasts were evaluated for 2021–2026, producing six forecast errors for each model.

For GPU demand, the initial training period covered 2022–2024, with forecasts evaluated for 2025 and 2026. This provided only two forecast errors, making GPU model comparisons less reliable than those

for overall HPC and CPU demand. Because evaluation used one-step-ahead forecasts, the accuracy measures do not directly establish performance over the full four-year projection horizon.

3.4.2 Forecast Accuracy Measures

Model accuracy was assessed using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), two commonly used measures of forecast accuracy (Hyndman and Koehler 2006; Hyndman and Athanasopoulos 2021).

RMSE was calculated as:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}$$

MAE was calculated as:

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|$$

where y_t is the observed value, $\hat{y}_t$ is the forecast value and n is the number of evaluated forecasts.

Both measures are expressed in the same units as the forecasting target. Lower values indicate better predictive accuracy.

RMSE gives greater weight to larger errors, while MAE measures the average absolute error. Both were considered because individual observations can have a substantial influence on accuracy measures when the evaluation period is short.

Model performance was compared within each demand measure. RMSE and MAE values were not compared directly across HPC, CPU and GPU series because their measurement units differ.

3.5 Model Selection and Sensitivity Analysis

Model selection considered predictive accuracy, interpretability and the assumptions required to extend historical trends through 2030.

A model with the lowest historical error was not automatically selected if its growth assumptions produced projections that were difficult to justify from the available evidence.

The linear and log-linear specifications were compared particularly closely because they represent different future growth patterns. Linear models assume a constant absolute annual increase, while log-linear models imply compounded proportional growth.

A sensitivity analysis was also conducted by fitting trend models with and without the 2026 observation. This examined how strongly the latest reported year influenced estimated growth and the resulting projections.

The comparison was intended to assess sensitivity to the historical sample rather than determine whether the 2026 observation represented a temporary or permanent change.

Forecast trajectories derived from different model specifications were treated as alternative assumptions, not formal statistical prediction intervals.

3.6 Field of Research Scenario Development

FoR scenarios were developed separately from the overall HPC, CPU and GPU forecasting models.

The available FoR data describe the distribution of allocated computing resources across research fields. They do not provide a consistent historical series of independently observed demand for every research field.

A top-down approach was therefore used. The selected overall HPC demand projection was distributed across the nine harmonised FoR groups using alternative assumptions about future allocation shares.

For research field i and forecast year t, the indicative allocation-share-based demand was calculated as:

$$\hat{D}_{i,t} = \hat{D}_t \times s_i$$

where $\hat{D}_t$ is the overall forecast HPC demand and s_i is the assumed share for research field i.

Three historical periods were examined to construct alternative share assumptions:

Table 5: Historical periods used to construct FoR allocation-share scenarios.

Scenario	Historical basis	Purpose
Recent composition	2021–2026	Reflect the more recent FoR reporting period
Full-period composition	2014–2026	Reflect the broader historical allocation pattern
Earlier composition	2014–2020	Examine sensitivity to the earlier research-field distribution

For each scenario, the allocation share of each FoR group was averaged across available years within the selected period, giving equal weight to each year. Missing group-year observations were not automatically assigned zero. The resulting average shares were normalised to sum to 100% before being applied to the overall HPC demand forecast.

These scenarios show how different historical allocation patterns affect the distribution of projected HPC demand. They are not independently validated forecasts of future requests by research field.

3.7 Forecast Horizon and Interpretation

Forecasts were generated for 2027–2030 by fitting the selected models to the available historical observations. Alternative growth assumptions and models excluding 2026 were retained for sensitivity analysis.

The projections extend historical demand trends rather than estimate future infrastructure capacity requirements directly. They do not explicitly account for changes in research policy, computing technology

or access to HPC resources. Historical accounting differences and the limited annual observations, particularly for GPU demand, also restrict forecast reliability.

4 Results

4.1 Initial Analysis

The initial analysis examined historical HPC demand and allocation, CPU and GPU resources, and the distribution of allocations across Fields of Research (FoR). The results were used to identify trends, assess differences in reporting coverage and determine which series were suitable for forecasting.

4.1.1 Historical HPC Demand and Allocation

Figure 1 shows historical HPC requests and allocations from 2014 to 2026.

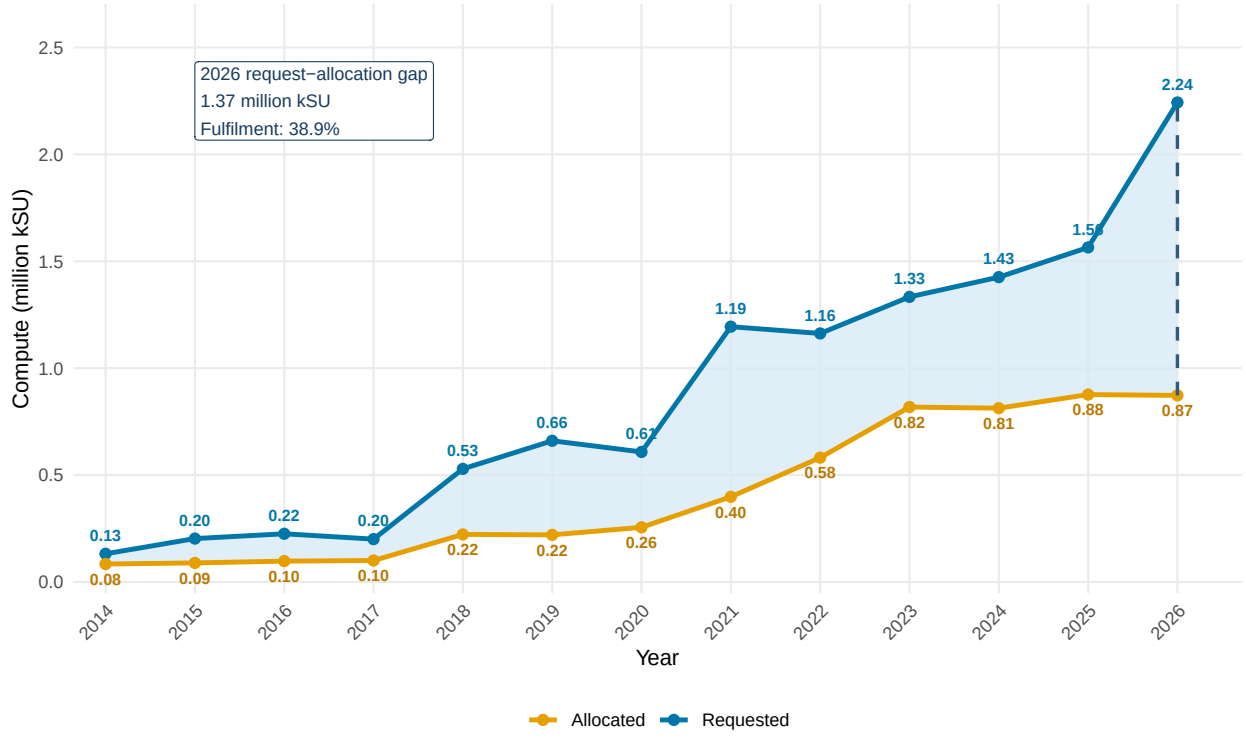


Figure 1: Historical HPC requests and allocations, 2014–2026. Shading represents the recorded request–allocation gap. Values are in million kSU.

Recorded requests increased from 131,356 kSU in 2014 to 2,242,817 kSU in 2026, while allocations increased from 83,505 to 872,288 kSU. Between 2023 and 2026, requests grew by approximately 68%, compared with 6% for allocations. The request–allocation gap reached 1.37 million kSU in 2026, and the fulfilment rate fell from 56.0% in 2025 to 38.9%.

The 2026 increase in requests was approximately 43.3% relative to the reconstructed 2025 value. Changes in computing systems and accounting practices may also affect historical comparisons, particularly

around the Raijin–Gadi and Magnus–Setonix transitions. The 2026 allocation figure excludes 12,000 kSU associated with Setonix Q. The observed gap indicates pressure on allocated resources, but does not measure all unmet national research demand.

4.1.2 CPU Demand and Allocation

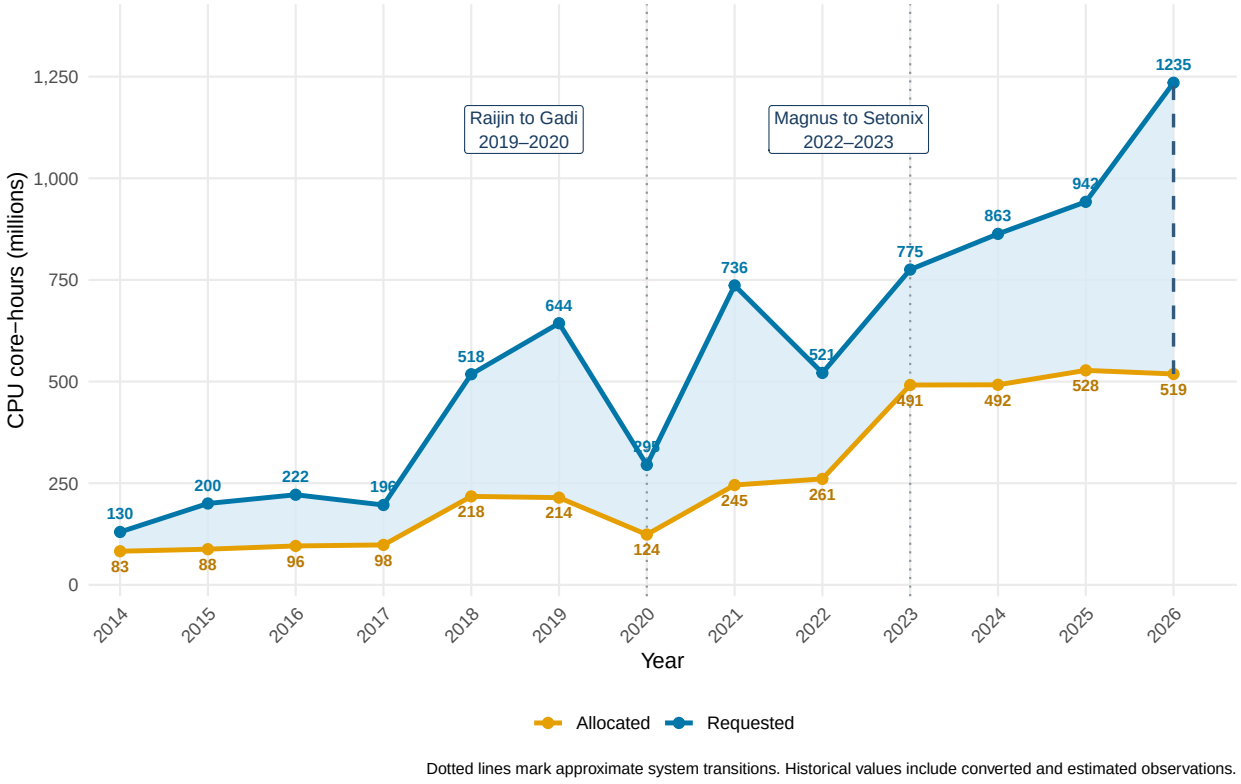


Figure 2: Historical CPU requests and allocations, 2014–2026. Shading shows the recorded gap; values include system-specific conversions and estimates.

Figure 2 shows historical CPU requests and allocations from 2014 to 2026, including converted and estimated values.

Estimated CPU requests reached approximately 1.24 billion core-hours in 2026, compared with allocations of approximately 519 million core-hours. Requests increased substantially after 2022, while allocations changed relatively little between 2023 and 2026.

The reconstructed series contains sharp changes around 2019–2022, when Raijin was replaced by Gadi and Magnus by Setonix. These transitions involved different computing systems and accounting rules, but the available records do not establish them as the causes of the fluctuations. Some 2019/20 observations also relied on a Gadi CPU-equivalent assumption because the CPU/GPU breakdown was unavailable. The long-term increase in estimated CPU requests is therefore more informative than individual year-to-year changes.

4.1.3 GPU Demand and Allocation

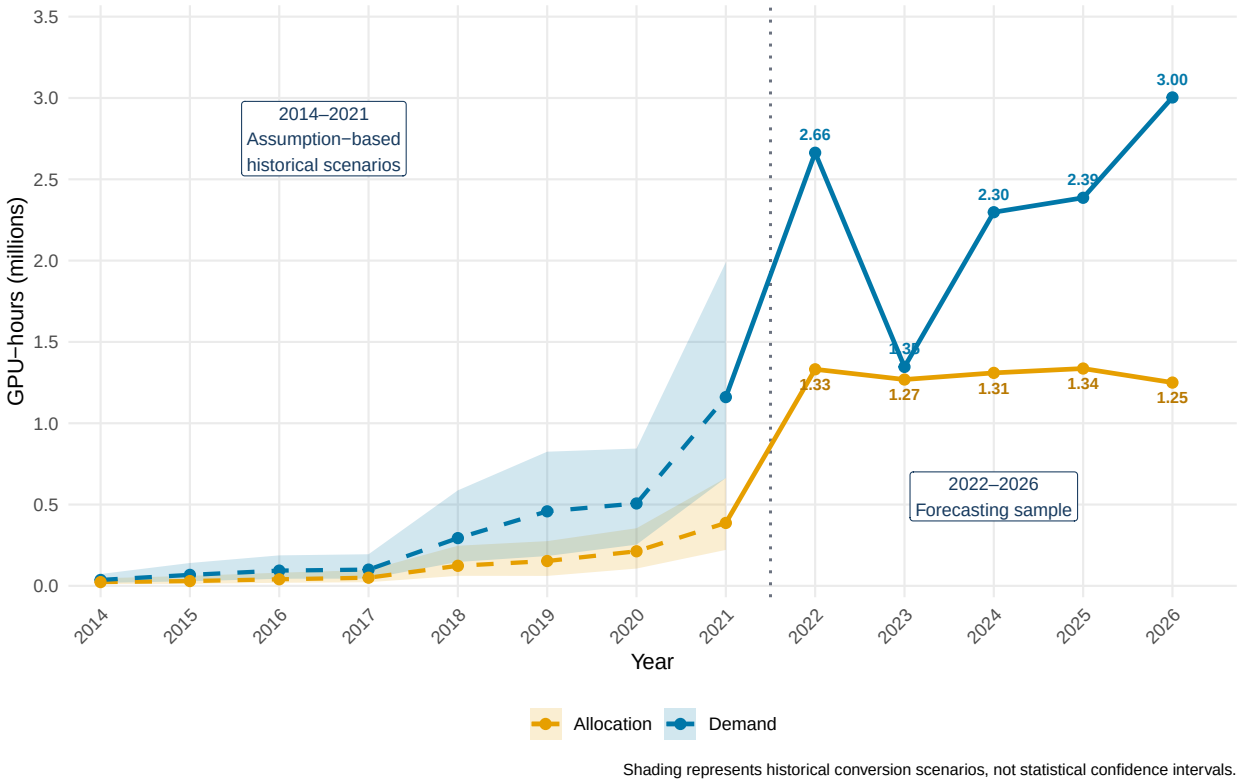


Figure 3: Historical GPU demand and allocation, 2014–2026. Dashed lines and shaded ranges indicate assumption-based historical estimates; solid lines show the 2022–2026 forecasting sample.

4.1.4 GPU Demand and Allocation

Figure 3 shows historical GPU requests and allocations, including alternative conversion assumptions for 2014–2021 and the shorter 2022–2026 forecasting sample.

Earlier GPU records lacked consistent GPU-hour measures and system-specific conversion information. Baseline and low–high estimates were therefore constructed using documented assumptions. The shaded range represents differences between these historical conversion assumptions, not statistical confidence intervals. Because these estimates depend heavily on the assumptions, they were excluded from model evaluation.

The 2022–2026 series provides a more consistent, although still partly estimated, basis for forecasting. Estimated GPU requests reached approximately 3.00 million GPU-hours in 2026, compared with allocations of approximately 1.25 million GPU-hours. Official Setonix GPU requests also increased by approximately 123% in service units between 2023 and 2026, although this percentage cannot be interpreted as equivalent growth in GPU-hours.

Estimated GPU requests increased from 2023 to 2026, but the five-year modelling sample and reliance on converted observations limit the reliability of forecasts through 2030.

4.1.5 Field of Research Allocation Composition

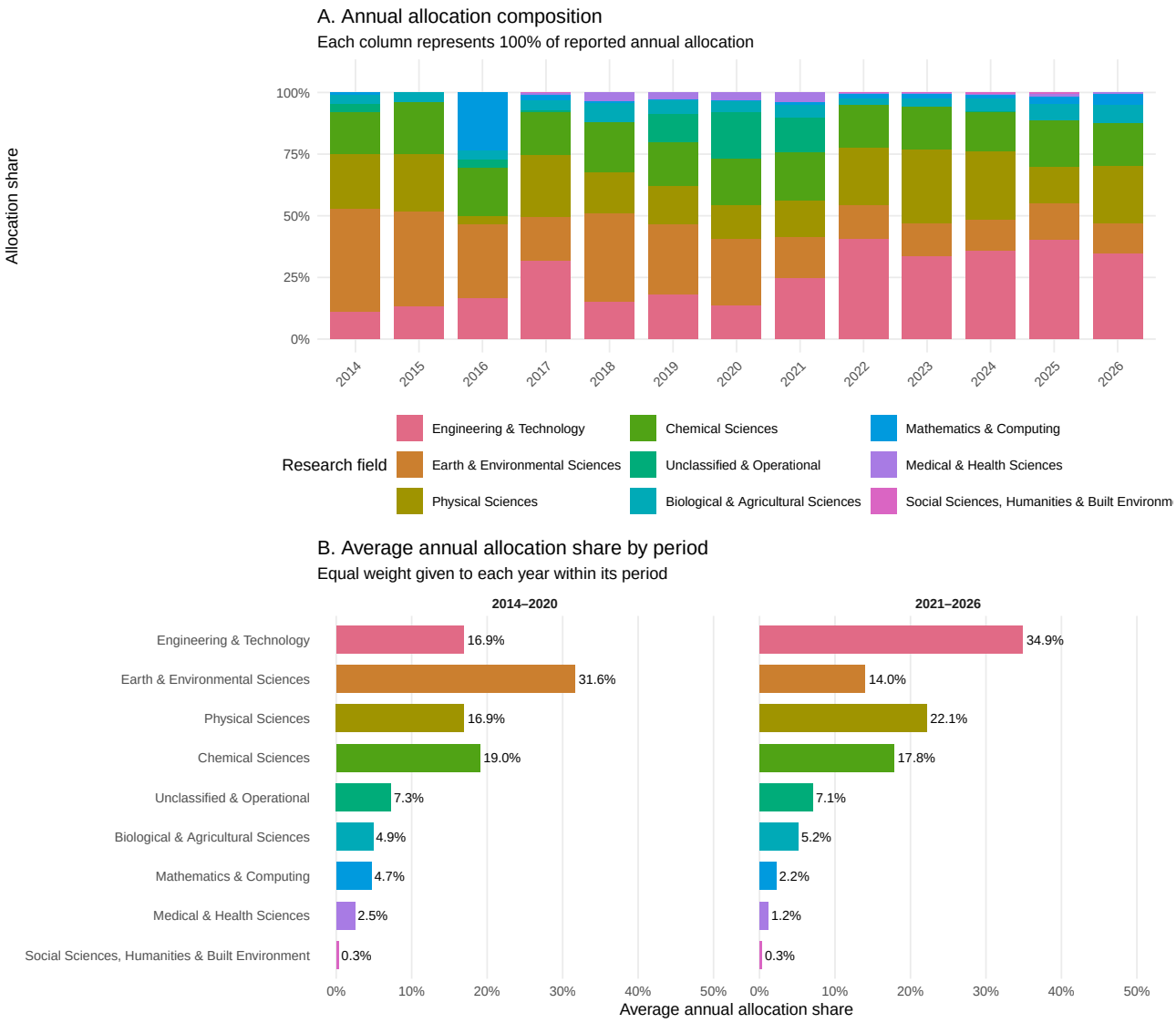


Figure 4: HPC allocation composition by Field of Research, 2014–2026. The lower panel compares average annual shares across two reporting periods.

Figure 4 shows the annual distribution of HPC allocations across nine analytical Field of Research (FoR) groups and compares average annual shares between 2014–2020 and 2021–2026.

During 2014–2020, Earth and Environmental Sciences accounted for approximately 32% of average annual allocations, followed by Chemical Sciences at 19%. Physical Sciences and Engineering and Technology each represented approximately 17%. In 2021–2026, Engineering and Technology became the largest group at approximately 35%, followed by Physical Sciences at 22%, Chemical Sciences at 18%, and Earth and Environmental Sciences at 14%.

These differences should be interpreted alongside the change from FoR 2008 to FoR 2020 classifications and the inclusion of Pawsey-wide reporting in 2016/17. The nine analytical groups support comparison

but cannot eliminate differences in classification or reporting scope.

The results describe allocation shares rather than independently observed demand by research field. The earlier and recent distributions are used to construct alternative FoR scenarios in Section 4.4.

4.2 Forecast Model Evaluation

The naïve, drift, linear trend and log-linear trend models were compared using one-year-ahead rolling-origin forecasts. Overall HPC and CPU accuracy was evaluated over 2021–2026, while GPU accuracy was evaluated over 2025–2026. Results are reported separately because the three demand measures use different units and historical periods.

4.2.1 Overall HPC Demand

Figure 5 compares observed HPC demand with one-year-ahead forecasts from the four models. The naïve model generally underestimated demand during periods of growth, while drift and linear trend followed the observed series more closely. The log-linear model tended to overestimate demand, particularly after 2022. However, all four models underestimated the sharp increase in 2026.

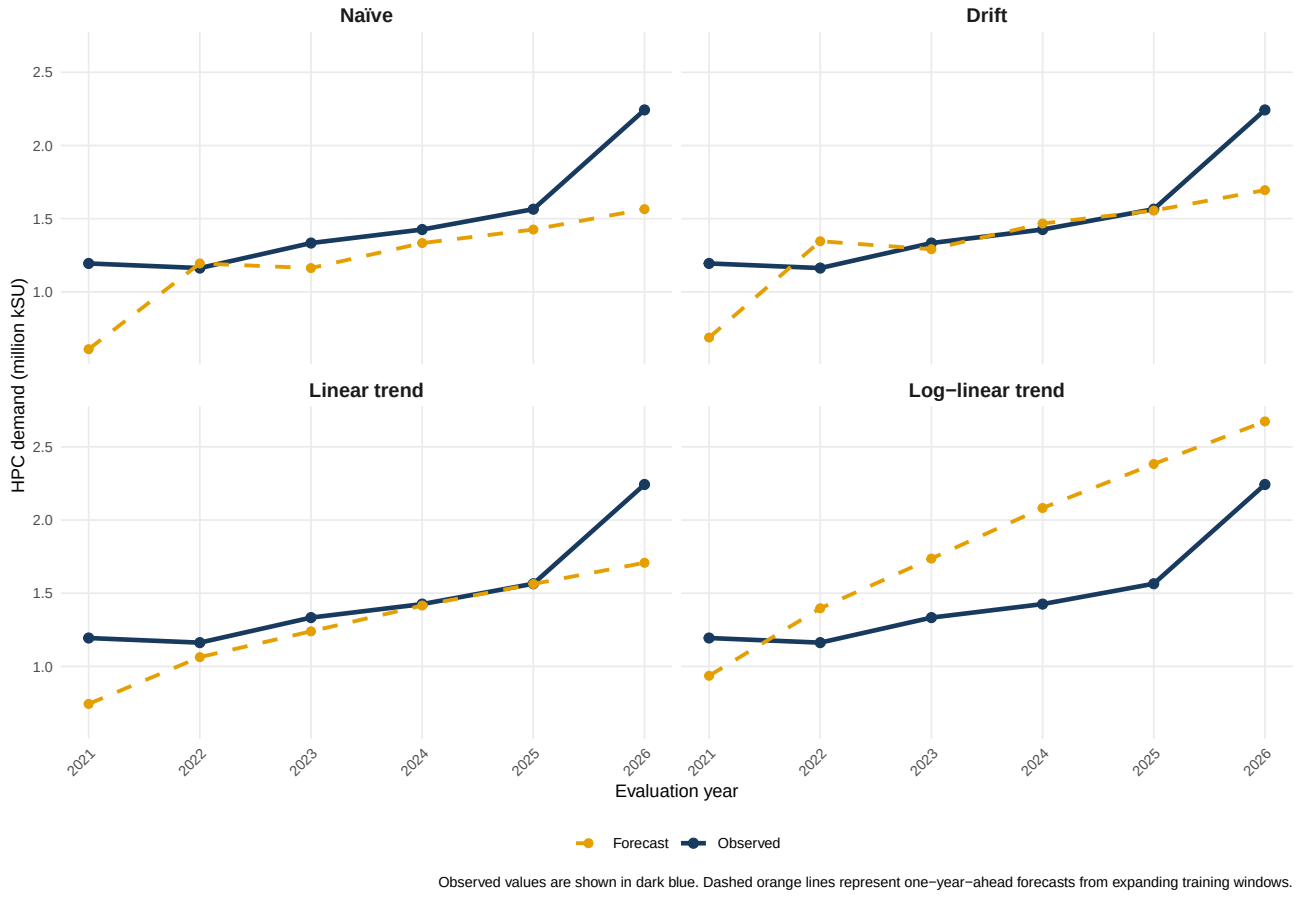


Figure 5: Observed overall HPC demand and rolling-origin one-year-ahead forecasts, 2021–2026. All panels use the same vertical scale.

Table 6 summarises forecast accuracy over 2021–2026. The linear trend achieved the lowest RMSE

(290,898 kSU) and MAE (198,295 kSU). Its RMSE was 23.2% lower than naïve and 7.5% lower than drift. The log-linear model performed worst, with an RMSE of 511,614 kSU.

Table 6: Rolling-origin forecast accuracy for overall HPC demand, 2021–2026.

Model	RMSE (kSU)	MAE (kSU)	RMSE rank
Linear trend	**290,898**	**198,295**	**1**
Drift	314,588	221,778	2
Naïve	378,864	283,000	3
Log-linear trend	511,614	466,991	4

The linear trend was selected as the central HPC forecasting model because it had the lowest evaluation errors and assumes a constant absolute annual increase. However, including 2026 increased the fitted annual trend from approximately 144,357 to 161,992 kSU (12.2%). The model excluding 2026 was therefore retained for sensitivity analysis, while the log-linear model was retained to examine continued proportional growth.

The evaluation is based on six one-year-ahead forecasts. It does not establish how accurately the selected model will predict demand through 2030. Section 4.3 compares the resulting projections and alternative growth assumptions.

4.2.2 CPU Demand

Figure 6 compares observed CPU demand with one-year-ahead forecasts from the four models over 2021–2026. The naïve and drift models struggled with changes in demand, particularly the decline in 2022. The linear trend produced forecasts relatively close to observed demand during 2023–2025 but underestimated demand in the later evaluation years. The log-linear model generally overestimated demand from 2022 to 2025, although its 2026 forecast was closer to the observed value.

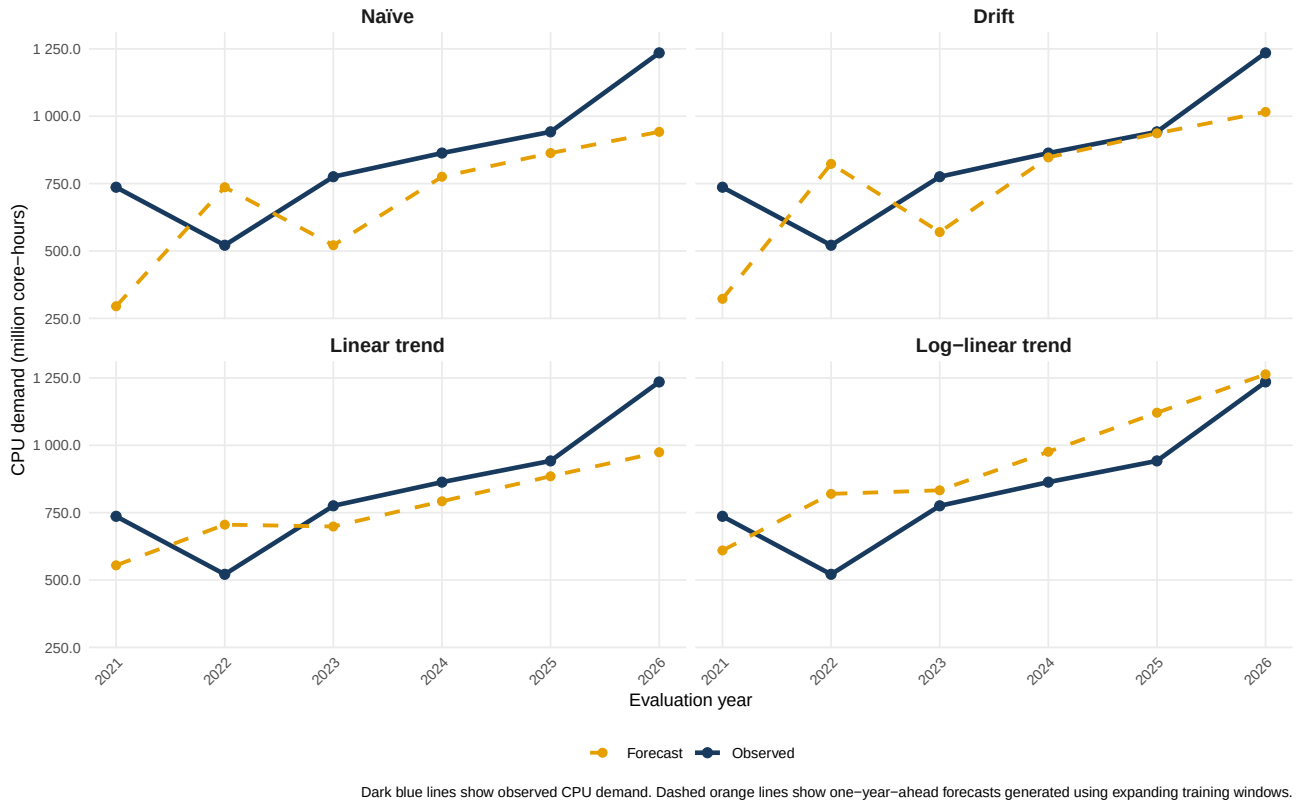


Figure 6: Observed CPU demand and rolling-origin one-year-ahead forecasts, 2021–2026. All panels use the same vertical scale.

Table 7 reports forecast accuracy across the six evaluation years. The linear trend achieved the lowest RMSE (157.72 million core-hours), while the log-linear model achieved the lowest MAE (133.74 million core-hours). The difference in RMSE between the two models was approximately 1.5%, compared with 3.5% for MAE. Neither model performed best on both measures.

Table 7: Rolling-origin forecast accuracy for CPU demand, 2021–2026.

Model	RMSE (million core-hours)	MAE (million core-hours)	RMSE rank
Linear trend	**157.72**	138.65	**1**
Log-linear trend	160.18	**133.74**	2
Drift	242.61	193.69	3
Naïve	260.05	228.45	4

The linear trend was selected as the central CPU forecasting model because it had the lowest RMSE and assumes a constant annual increase in demand. The log-linear model was retained as an alternative because its MAE was lower and its proportional-growth assumption produces a different forecast trajectory through 2030.

Including the 2026 observation increased the estimated annual growth rate in the linear model from 72.41 million to 81.01 million CPU core-hours (11.9%). This shows that the fitted trend is sensitive to

the latest observation. The model excluding 2026 was retained to assess how this difference affects the projections through 2030 (Section 4.3).

4.2.3 GPU Demand

Figure 7 compares observed GPU demand with one-year-ahead forecasts for 2025 and 2026. The vertical segments show the forecast errors for each model. All four models underestimated demand in 2026, when estimated GPU requests reached approximately 3.00 million GPU-hours. The naïve model forecast approximately 2.39 million GPU-hours, while the trend models also failed to anticipate the increase.

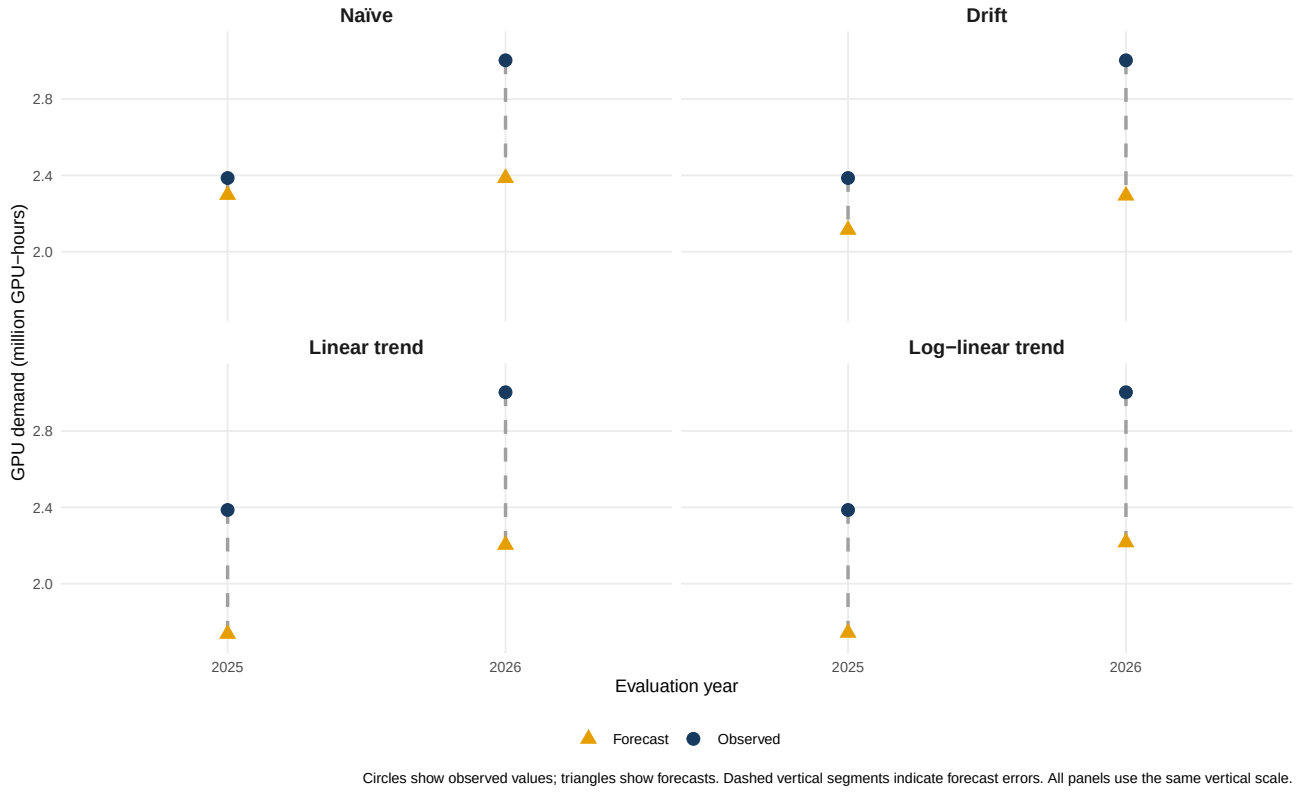


Figure 7: Observed GPU demand and exploratory one-year-ahead forecasts, 2025–2026. Vertical segments show the difference between observed and forecast values. Only two evaluation years were available.

Table 8 reports the accuracy results from the two evaluation years. The naïve model achieved the lowest RMSE (0.441 million GPU-hours) and MAE (0.353 million GPU-hours), followed by drift, log-linear trend and linear trend.

Table 8: Exploratory rolling-origin forecast accuracy for GPU demand, 2025–2026.

Model	RMSE (million GPU-hours)	MAE (million GPU-hours)	RMSE rank
Naïve	0.441	0.353	1
Drift	0.537	0.490	2
Log-linear trend	0.719	0.715	3

Table 8: Exploratory rolling-origin forecast accuracy for GPU demand, 2025–2026.

Model	RMSE (million GPU-hours)	MAE (million GPU-hours)	RMSE rank
Linear trend	0.729	0.725	4

These rankings are based on only two forecast errors per model. The GPU series also includes converted and estimated observations, so the results do not establish which method would perform best over a longer period. In particular, the naïve model's lower evaluation errors do not imply that GPU demand will remain constant through 2030.

The naïve model was retained as a reference, while linear and log-linear trends were used to examine alternative growth assumptions. No model was selected as a central GPU forecast. GPU projections were therefore treated as exploratory and excluded from the initial national HPC planning scenarios, which use the overall HPC demand forecast and historical Field of Research allocation shares.

Section 4.3 presents the exploratory GPU projections through 2030.

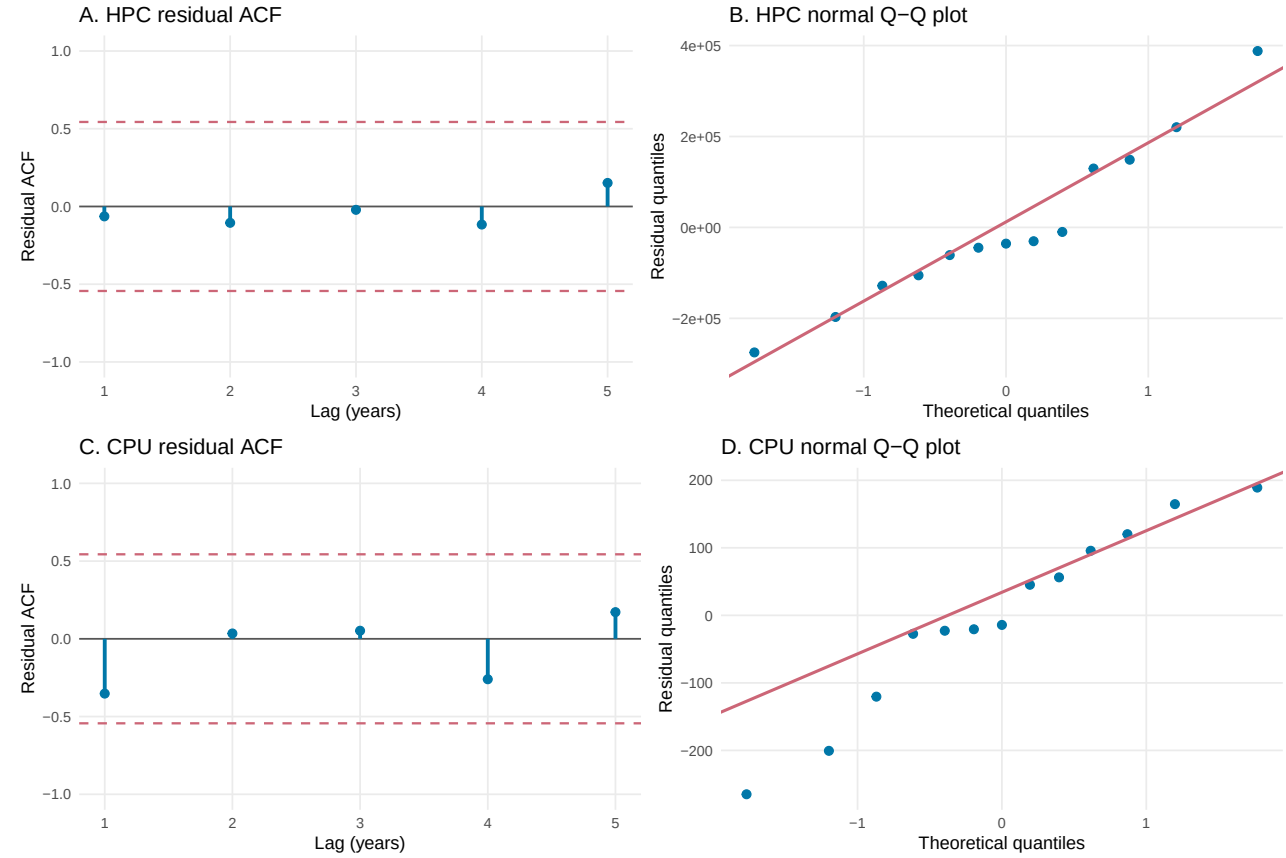
4.2.4 Residual Diagnostics

The rolling-origin evaluation assessed how accurately the forecasting models predicted observations excluded from their training samples. Residual diagnostics were then conducted on the selected linear trend models for overall HPC and CPU demand, using the complete 2014–2026 historical series.

The purpose was to examine whether the fitted models left systematic patterns in their errors and whether their residual distributions supported the assumptions used to construct statistical prediction intervals.

Residual autocorrelation was examined using the autocorrelation function (ACF) and the Ljung–Box test. Normal Q–Q plots were also examined to assess departures from the normal-error assumption. These diagnostics complement the out-of-sample evaluation presented in Sections 4.2.1 and 4.2.2.

Figure Figure 8 presents the residual diagnostics for both selected models.



stics use in−sample residuals from linear trend models fitted to 2014–2026. Dashed ACF lines are approximate reference limits. CPU residuals are measured in million core−hours.

Figure 8: Residual diagnostics for the selected linear trend models fitted to 2014–2026: residual ACF and normal Q–Q plots for overall HPC and CPU demand.

The residual ACF plots show relatively small correlations for both models. None of the displayed correlations exceeds the approximate reference limits.

For overall HPC demand, the residual correlations are close to zero across most displayed lags. This suggests that the linear trend captures much of the systematic movement in the historical series, although the short sample limits the strength of this conclusion.

CPU residuals show a more noticeable negative correlation at lag 1 and lag 4. These correlations remain within the approximate reference limits, but they indicate that some year-to-year variation may not be fully captured by a single linear trend.

The Ljung–Box test was used to examine whether the residual autocorrelations were jointly distinguishable from zero over the first three lags. Table Table 9 reports the results.

Table 9: Ljung–Box tests of residual autocorrelation for the selected linear trend models, using three lags.

Demand measure	Ljung-Box statistic	Lag	Degrees of freedom	p-value
Overall HPC	0.271	3	3	0.965

Table 9: Ljung–Box tests of residual autocorrelation for the selected linear trend models, using three lags.

Demand measure	Ljung-Box statistic	Lag	Degrees of freedom	p-value
CPU	2.092	3	3	0.554

The Ljung–Box results do not provide sufficient evidence to reject the null hypothesis of no residual autocorrelation at the tested lags for either selected model, using a 5% significance level.

However, the results should not be interpreted as proof that the residuals are independent. Each model was fitted using only 13 annual observations, and the test has limited power with such a small sample. The ACF reference limits are also approximate.

The normal Q–Q plots provide additional information about the distribution of the fitted errors.

For overall HPC demand, most residuals lie reasonably close to the theoretical reference line, although some departures are visible around the centre and upper tail. The plot does not show an obvious severe departure from normality, but the small sample makes the distribution difficult to assess reliably.

The CPU Q–Q plot shows more noticeable departures, particularly in the lower tail. The most negative residual falls substantially below the reference line, suggesting that the normal-error assumption may be less suitable for CPU demand.

These differences matter when interpreting model-based prediction intervals. The intervals depend on assumptions about the fitted error distribution, and departures from those assumptions may affect their coverage. The CPU intervals therefore require particular caution.

Overall, the diagnostics do not reveal strong evidence of residual autocorrelation in the selected linear models. However, they do not establish that the fitted trends or their error distributions will remain unchanged through 2030.

The residual diagnostics support the use of the linear models as interpretable forecasting references, while reinforcing the need to consider alternative growth assumptions and sensitivity to the 2026 observation. These issues are examined in Section 4.3.

4.2.5 Evaluation Summary

The rolling-origin evaluation identified different forecasting results for overall HPC, CPU and GPU demand. The model selection considered forecast accuracy, historical data coverage, growth assumptions and the reliability of the underlying observations.

For **overall HPC demand**, the linear trend achieved the lowest RMSE (290,898 kSU) and MAE (198,295 kSU) across six evaluation years. Its RMSE was 23.2% lower than the naïve benchmark, although the improvement over drift was smaller at 7.5%. The linear model was therefore selected as the central forecasting model, with drift providing a useful benchmark. The log-linear model was retained as an alternative growth scenario despite its weaker historical accuracy.

For **CPU demand**, the results were less decisive. The linear trend achieved the lowest RMSE (157.72 million core-hours), while the log-linear model achieved the lowest MAE (133.74 million core-hours).

The difference in RMSE between the two models was only 1.5%. The linear model was selected for the central projection because it provided competitive accuracy and assumed a constant annual increase rather than continued proportional growth. The log-linear model was retained to examine how this alternative assumption affects demand through 2030.

Both selected linear models were sensitive to the latest observation. Including 2026 increased the estimated annual growth by 12.2% for overall HPC demand and 11.9% for CPU demand. This supported retaining forecasts excluding 2026 as sensitivity comparisons, rather than relying only on models fitted to the complete historical series.

Residual diagnostics provided no strong evidence of autocorrelation in either selected linear model. However, the Ljung–Box tests had limited power with 13 annual observations. The CPU Q–Q plot also showed departures from normality, particularly in the lower tail. These findings indicate that statistical prediction intervals should be interpreted cautiously, especially for CPU demand.

For **GPU demand**, the naïve benchmark achieved the lowest RMSE (0.441 million GPU-hours) and MAE (0.353 million GPU-hours). However, the evaluation used only two forecast years, and the underlying GPU-hour series depended partly on conversion assumptions. These results were insufficient to select a reliable long-term forecasting model. Following the project review, GPU projections were retained for exploratory analysis but excluded from the initial national HPC planning scenarios.

Overall, the evaluation supports using the **linear trend as the central reference for overall HPC and CPU demand**, while retaining alternative growth assumptions and models excluding 2026 to examine forecast sensitivity. The initial national HPC planning scenarios are based on the overall HPC projection and historical Field of Research allocation shares. Section 4.3 presents these projections and examines how the modelling assumptions affect the results through 2030.

4.3 Demand Forecasts to 2030

4.3.1 Overall HPC Demand

The rolling-origin evaluation in Section 4.2.1 identified the linear trend as the most accurate of the four models for overall HPC demand. The model was therefore fitted to the complete 2014–2026 series to generate the central projections for 2027–2030.

Three alternative specifications were retained to examine how the projections depend on the historical observations and growth assumptions. The first excludes 2026 from the linear model to assess the influence of the latest observation. The other two use log-linear trends, fitted with and without 2026, to examine the implications of continued proportional growth.

This comparison is important because recorded HPC requests increased sharply in 2026. Including this observation increased the estimated annual linear growth from approximately 144,357 to 161,992 kSU, or 12.2%. The alternative specifications therefore assess the sensitivity of the 2030 projections to both the latest observation and the assumed growth pattern.

Figure 9 connects the historical observations with the four projected trajectories. The 2030 values are labelled to show the differences between the forecasting assumptions.

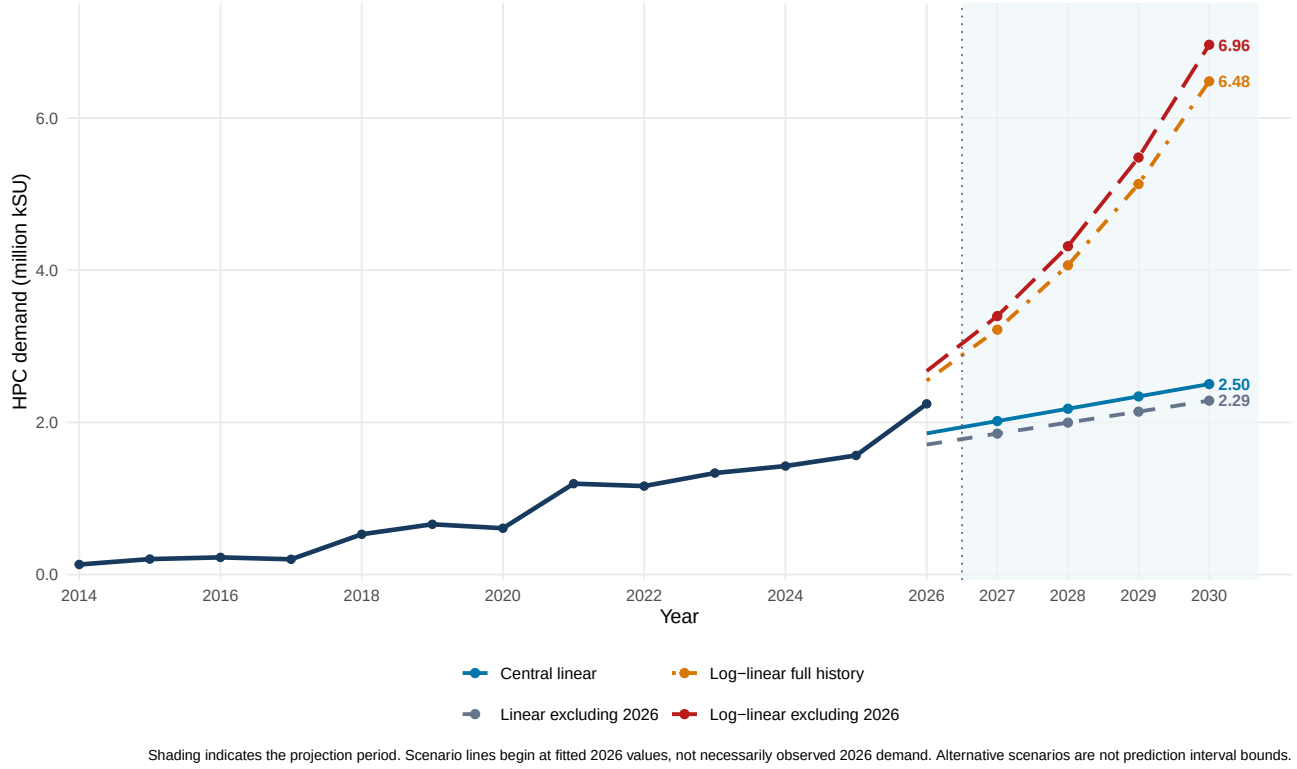


Figure 9: Historical HPC demand and alternative projections to 2030. The full-history linear model provides the central forecast. Alternative trajectories show sensitivity to the 2026 observation and growth assumptions.

The central linear model projects approximately 2.02 million kSU in 2027, increasing by around 161,992 kSU annually to reach 2.50 million kSU in 2030. This represents an increase of approximately 11.6% from the reconstructed 2026 demand of 2.24 million kSU.

Excluding the 2026 observation reduces the 2030 projection to approximately 2.29 million kSU, around 210,000 kSU below the central forecast. This difference shows that the latest observation influences the estimated growth rate, even when the same linear model is used.

The full-history log-linear model produces a substantially higher 2030 projection of approximately 6.48 million kSU. Unlike the linear model, which assumes a constant absolute annual increase, the log-linear model extends proportional growth. As a result, its projected annual increases become progressively larger.

However, the log-linear model recorded the highest RMSE and MAE during the rolling-origin evaluation in Section 4.2.1. Its higher projection is therefore retained to illustrate sensitivity to the growth assumption, rather than as an equally supported central forecast or a validated upper limit.

For planning purposes, the difference between the linear and log-linear projections is more substantial than the difference caused by excluding 2026. This indicates that the assumed growth pattern has a greater influence on the 2030 estimate than the latest observation alone. The linear model is therefore retained as the central reference, while the alternative trajectories illustrate how different assumptions affect the longer-term outlook.

These projections describe recorded HPC requests rather than unconstrained researcher demand. They provide a basis for considering future national computing requirements, but should not be interpreted as precise estimates of the capacity that facilities will need to provide. The statistical uncertainty around the selected central forecast is examined separately in Section 4.3.2.

4.3.1.1 Prediction Intervals and Forecast Uncertainty

The differences between forecasting scenarios describe uncertainty arising from model selection and historical data. Statistical prediction intervals address a different question: how much future annual demand might vary around the selected linear trend if the fitted model and its error assumptions remain appropriate.

Figure Figure 10 shows the central projection with 80% and 95% model-based prediction intervals.

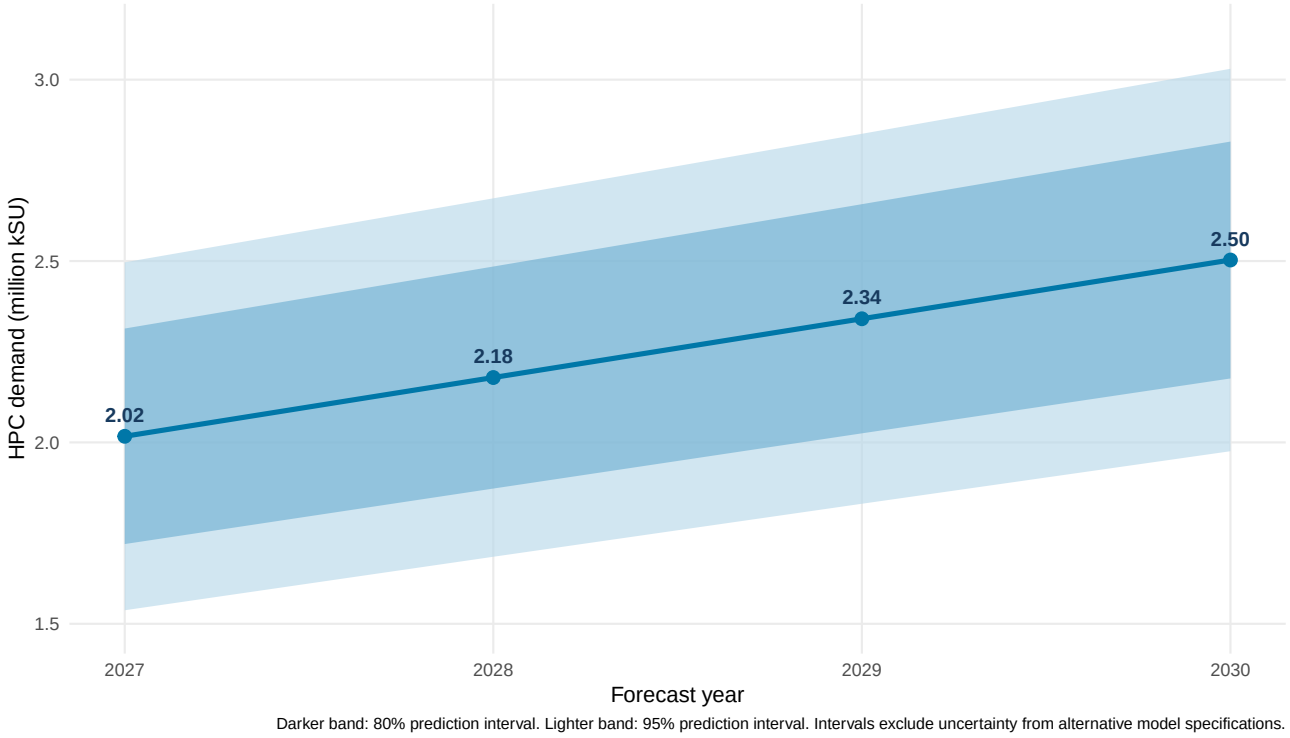


Figure 10: Central HPC demand projection with 80% and 95% model-based prediction intervals, 2027–2030. Intervals are conditional on the fitted linear model.

The selected linear model projects approximately 2.50 million kSU in 2030, with a 95% prediction interval of approximately 1.98–3.03 million kSU. The interval is substantially wider than the annual increase estimated by the model, showing that uncertainty remains important even when the central trend is relatively stable.

These intervals depend on the linear regression assumptions and the historical series used to fit the model. They do not account for possible changes in computing technology, reporting practices, research funding or the underlying relationship between requested and actual computing requirements.

The residual diagnostics in Section 4.2.4 provide some evidence about the fitted model's assumptions,

but the short annual series limits how confidently these assumptions can be extended through 2030.

Overall, the linear model provides a transparent central estimate of approximately 2.50 million kSU for national HPC planning in 2030. The model excluding 2026 demonstrates sensitivity to the latest observation, while the log-linear alternatives show how different growth assumptions can produce much larger projections. These results should be interpreted as conditional planning estimates rather than precise future capacity requirements.

4.3.2 CPU Demand

The rolling-origin evaluation in Section 4.2.2 showed that the linear and log-linear models performed similarly when forecasting CPU demand. The linear model achieved the lowest RMSE of 157.72 million core-hours, while the log-linear model achieved the lowest MAE of 133.74 million core-hours. The linear trend was selected as the central model because it provided competitive accuracy without assuming continued compounded growth.

The selected model was fitted to the complete 2014–2026 CPU series to generate projections for 2027–2030. Three alternative specifications were retained: a linear model excluding 2026, and log-linear models fitted with and without 2026. These alternatives examine the effects of the latest observation and different growth assumptions.

The 2026 observation is important because including it increased the estimated annual linear growth from approximately 72.41 million to 81.01 million core-hours, or 11.9%. With only 13 annual observations, the latest data can influence the estimated trend and projections several years ahead. Figure 11 compares the historical CPU demand series with the four projected trajectories.

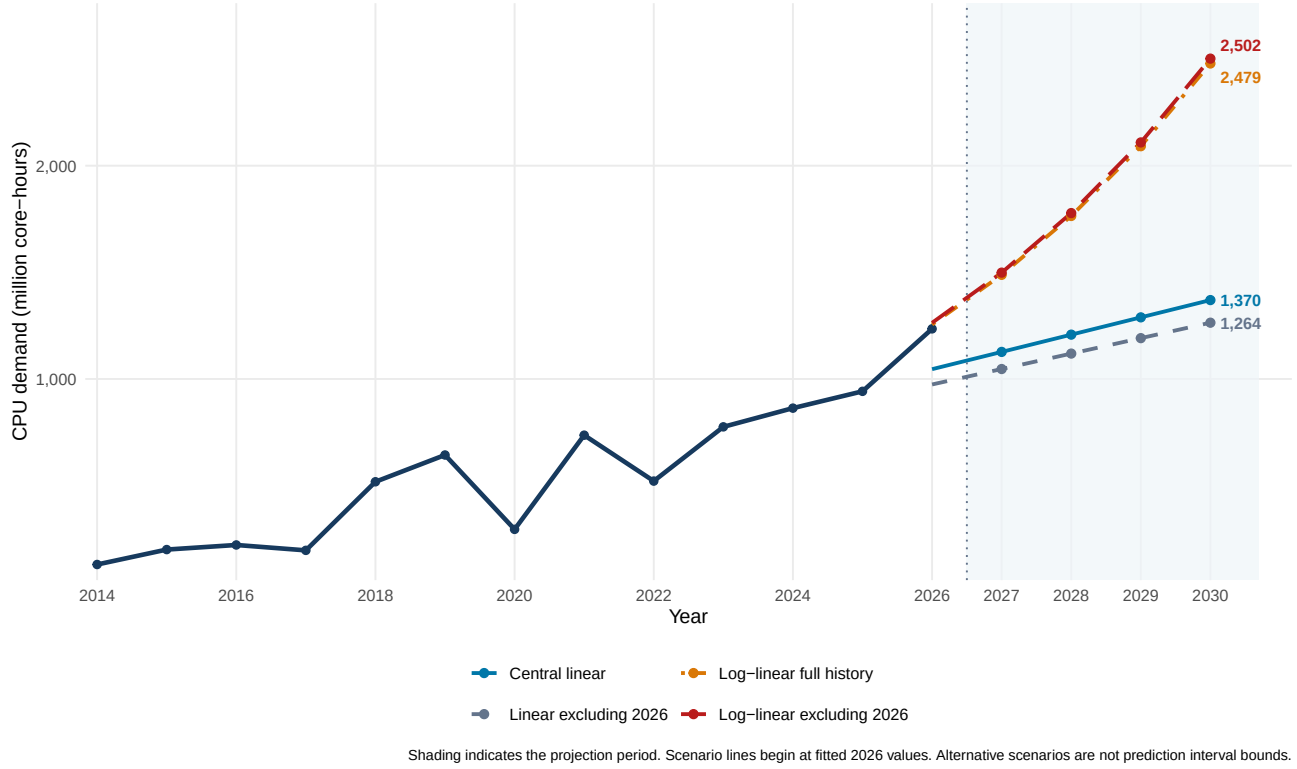


Figure 11: Historical CPU demand and alternative projections to 2030. The full-history linear model provides the central forecast; other trajectories illustrate sensitivity to the 2026 observation and growth assumptions.

The central linear model projects CPU demand of approximately 1.13 billion core-hours in 2027, increasing by around 81.01 million core-hours annually to reach 1.37 billion core-hours in 2030. This represents an increase of approximately 10.9% from the reconstructed 2026 CPU demand.

Excluding the 2026 observation reduces the 2030 projection to approximately 1.26 billion core-hours, around 106 million core-hours below the central forecast. This difference indicates that the latest observation has a noticeable influence on the fitted trend. With only 13 annual observations, changes in the latest data can affect projections several years ahead.

The full-history log-linear model produces a substantially higher projection of approximately 2.48 billion core-hours in 2030. This is approximately 81% higher than the central linear projection for the same year, despite the two models achieving similar accuracy during the one-year-ahead evaluation period. The difference arises because the log-linear model extends proportional growth, while the linear model assumes a constant absolute annual increase.

The two log-linear specifications also produce relatively similar projections when 2026 is included or excluded. This contrasts with the linear models, where excluding 2026 has a more noticeable effect on the 2030 projection. The comparison suggests that the log-linear CPU projections are less sensitive to the latest observation, although both depend on the assumption of continued proportional growth.

Unlike overall HPC demand, the log-linear CPU model performed competitively during rolling-origin evaluation. Its RMSE was approximately 1.5% higher than the linear model, while its MAE was

approximately 3.5% lower. Therefore, the higher log-linear projection cannot be dismissed on the basis of historical forecast accuracy alone.

However, similar one-year-ahead accuracy does not establish that proportional growth will continue through 2030. The difference between the linear and log-linear projections illustrates how model assumptions become increasingly important over longer forecast horizons.

The linear model was retained as the central planning reference because it achieved the lowest RMSE and provides a more conservative and interpretable projection. The log-linear alternatives were retained to examine the implications of continued proportional growth rather than treated as equally likely outcomes.

4.3.2.1 CPU Prediction Intervals

The alternative CPU projections illustrate uncertainty associated with model selection and the inclusion of the 2026 observation. Prediction intervals address a different source of uncertainty by estimating how future observations may vary around the selected linear trend, assuming its statistical assumptions remain appropriate.

Figure 12 presents the central CPU forecast with 80% and 95% prediction intervals for 2027–2030.

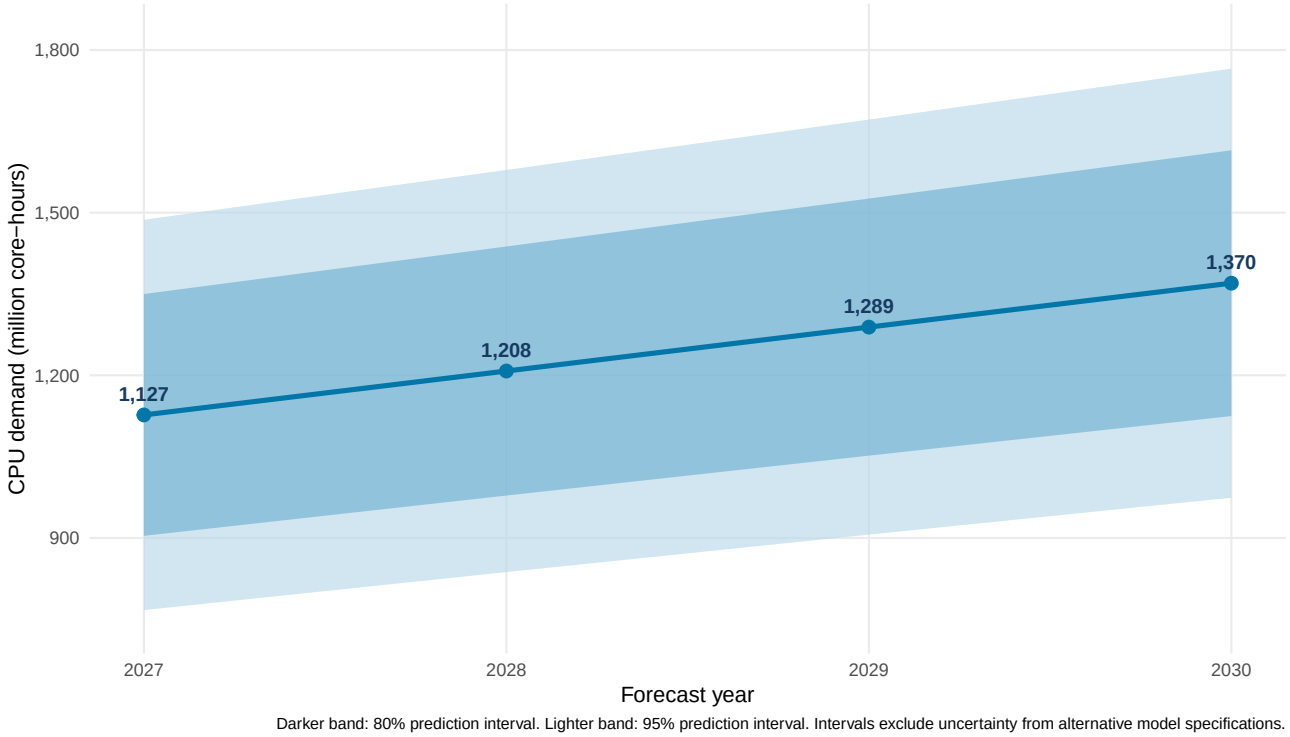


Figure 12: Central CPU demand projection with 80% and 95% model-based prediction intervals, 2027–2030. Intervals are conditional on the fitted linear model.

The selected linear model projects approximately 1,369.90 million CPU core-hours in 2030, with a 95% prediction interval of approximately 974.24–1,765.55 million core-hours. The interval is relatively wide compared with the estimated annual increase of 81.01 million core-hours, indicating substantial

uncertainty around individual future observations.

These intervals are conditional on the fitted linear model. They do not account for additional uncertainty associated with historical resource conversions, changes in computing infrastructure, reporting practices or alternative growth assumptions.

The full-history log-linear projection of approximately 2.48 billion core-hours lies well above the central model's 95% prediction interval. This does not invalidate either model. Instead, it demonstrates that a prediction interval calculated from one model does not capture the different outcomes produced by alternative model specifications.

The residual diagnostics in Section 4.2.4 provide additional information about the fitted model assumptions. However, the limited number of annual observations and historical reporting differences mean that the statistical intervals should be interpreted cautiously.

The central CPU forecast provides a reference for assessing future computing demand, while the alternative trajectories show how projections change under different growth assumptions. The large difference between the linear and log-linear projections indicates that planning decisions should consider more than one possible growth path. These CPU estimates should also be interpreted alongside overall HPC demand, rather than added together, because they represent related measures of computing requirements.

4.3.3 GPU Demand

Unlike overall HPC and CPU demand, no central GPU forecasting model was selected. Formal prediction intervals were not reported because the five-year modelling series and uncertainty in the underlying estimates provided insufficient evidence for reliable statistical intervals.

Following the project review, GPU projections were retained as exploratory findings and excluded from the initial national HPC planning scenarios. Those scenarios instead use the selected overall HPC demand projection and historical Field of Research allocation shares.

The GPU results nevertheless highlight an important limitation for future planning. More consistent records of GPU requests, allocations, usage and hardware configurations would be needed to support reliable GPU-specific forecasts. Until such data become available, the projections should be used to examine modelling sensitivity rather than estimate future national GPU capacity requirements.

Figure Figure 13 shows the historical GPU demand estimates and the five exploratory trajectories through 2030.

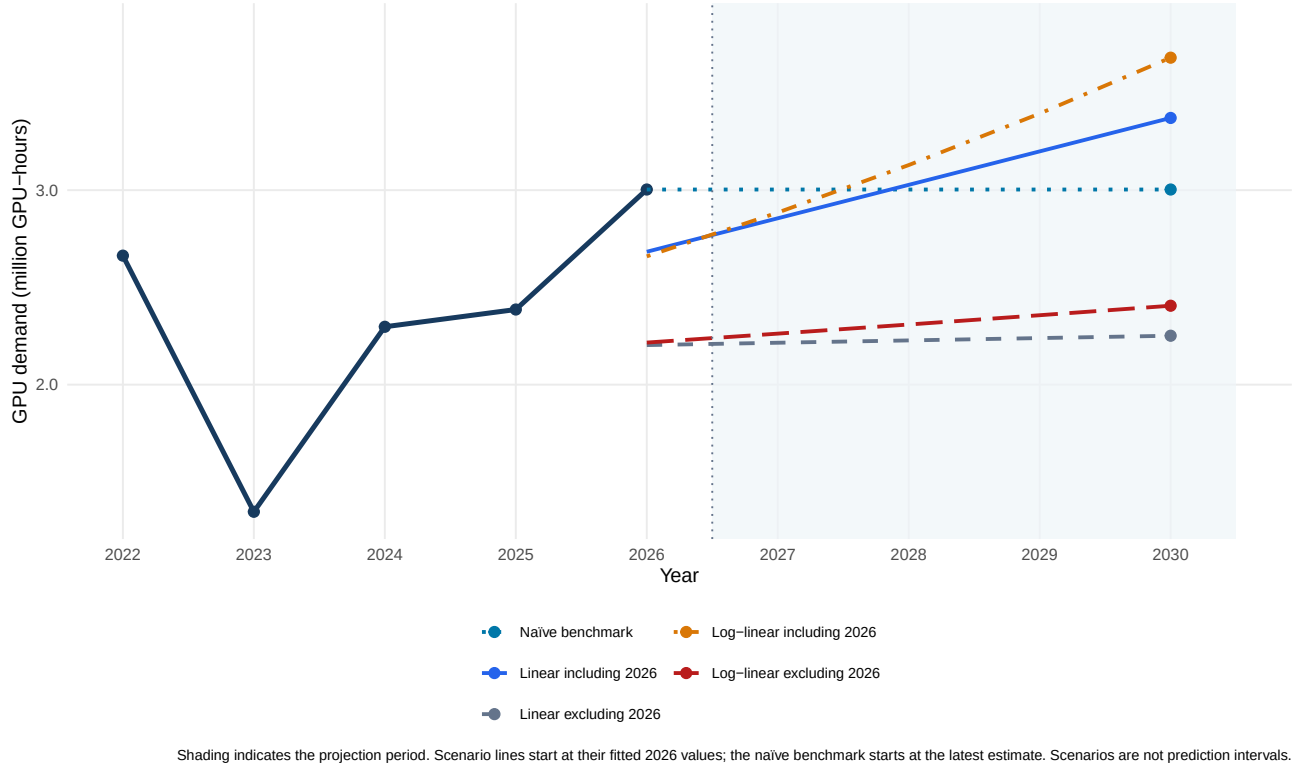


Figure 13: Historical GPU demand estimates and five exploratory projections to 2030. The scenarios show sensitivity to growth assumptions and the inclusion of 2026; they are not validated forecasts or prediction interval bounds.

The five scenarios produce different estimates of GPU demand in 2030. The naïve benchmark maintains the latest estimated demand at approximately 3.00 million GPU-hours. The full-history linear model projects 3.37 million GPU-hours, while the full-history log-linear model projects 3.68 million GPU-hours.

Excluding 2026 produces substantially lower projections. The linear model excluding 2026 projects approximately 2.25 million GPU-hours in 2030, while the corresponding log-linear model projects 2.41 million GPU-hours.

The difference between the two linear specifications is approximately 1.12 million GPU-hours in 2030. The full-history linear projection is around 49.7% higher than the projection excluding 2026. This shows that the latest observation has a substantial influence on the estimated growth trajectory.

Across the five scenarios, the 2030 estimates range from approximately 2.25 to 3.68 million GPU-hours. This range reflects differences in the historical sample and model assumptions. It should not be interpreted as a statistical prediction interval or as validated lower and upper limits for future GPU requirements.

The naïve benchmark's favourable rolling-origin performance also requires caution. Although it achieved the lowest errors over the two evaluation years, this does not establish that GPU demand will remain constant through 2030. Similarly, the higher projections from the trend models do not provide reliable evidence that continued growth will occur.

Unlike overall HPC and CPU demand, no central GPU forecasting model was selected. Formal 80%

and 95% prediction intervals were also not reported because the short historical series and uncertainty in the underlying estimates would make these intervals highly dependent on the modelling assumptions.

Following the project review, GPU projections were retained as exploratory findings but excluded from the initial national HPC planning scenarios. Those initial scenarios instead use the selected overall HPC demand projection and historical Field of Research allocation shares.

This decision reflects the limitations of the available data rather than the importance of GPU computing. More consistent records of GPU requests, allocations, usage and hardware configurations would be needed before GPU-specific forecasts could support infrastructure planning with greater confidence.

Overall, the GPU analysis demonstrates that different growth assumptions and the inclusion of a single recent observation can substantially change longer-term projections. The results are useful for identifying uncertainty and informing future modelling, but they should not be treated as validated estimates of national GPU capacity requirements.

4.4 Field of Research Scenarios

The previous sections presented national HPC and CPU demand projections, alongside exploratory GPU projections, through 2030. However, the national forecast does not show how computing demand might be distributed across research disciplines. This section examines that question using historical Field of Research (FoR) allocation shares.

The available FoR records do not provide a consistent series of demand requests for every research field. Therefore, the scenarios apply historical allocation shares to the selected national HPC demand projection. The results are illustrative distributions of projected national demand, not independently estimated forecasts of demand within individual research fields.

4.4.1 Scenario Construction

Three historical reference periods were considered:

1. **Earlier period (2014–2020):** The allocation composition recorded before 2021.
2. **Recent period (2021–2026):** The latest six years of allocation records.
3. **Full period (2014–2026):** The allocation composition across the complete historical series.

The scenario construction followed four steps. First, reported FoR categories were consolidated into nine analytical groups, as described in Section 4.1. Second, annual allocation shares were calculated for each reported group. Third, the mean of the available annual shares was calculated within each reference period and normalised to produce weights totalling 100%. Finally, these weights were applied to the selected national HPC demand projections for 2027–2030.

The calculation is:

$$D_{f,t}^{(s)} = \widehat{D}_t \times w_f^{(s)}$$

where $D_{f,t}^{(s)}$ represents the illustrative demand for research field f in year t under scenario s, $\widehat{D}_t$ is the national HPC demand projection, and $w_f^{(s)}$ is the normalised historical allocation share.

The normalisation is calculated as:

$$w_f^{(s)} = \frac{\overline{a}_f^{(s)}}{\sum_j \overline{a}_j^{(s)}}$$

where $\overline{a}_f^{(s)}$ is the mean observed annual allocation share for research field f during reference period s. The denominator sums the mean shares across the nine groups.

This method gives equal weight to each available annual observation within a research field, rather than giving more weight to years with larger total allocations. Missing categories are not treated as confirmed zero allocations. Normalisation ensures that the weights sum to 100%, but it does not correct incomplete historical reporting.

The weights are held constant throughout 2027–2030. This isolates the effect of choosing different historical allocation compositions without introducing additional assumptions about future changes in research-field shares.

4.4.2 Historical Allocation Assumptions

Figure 14 compares the normalised allocation shares for the three reference periods. All nine research-field groups are included.

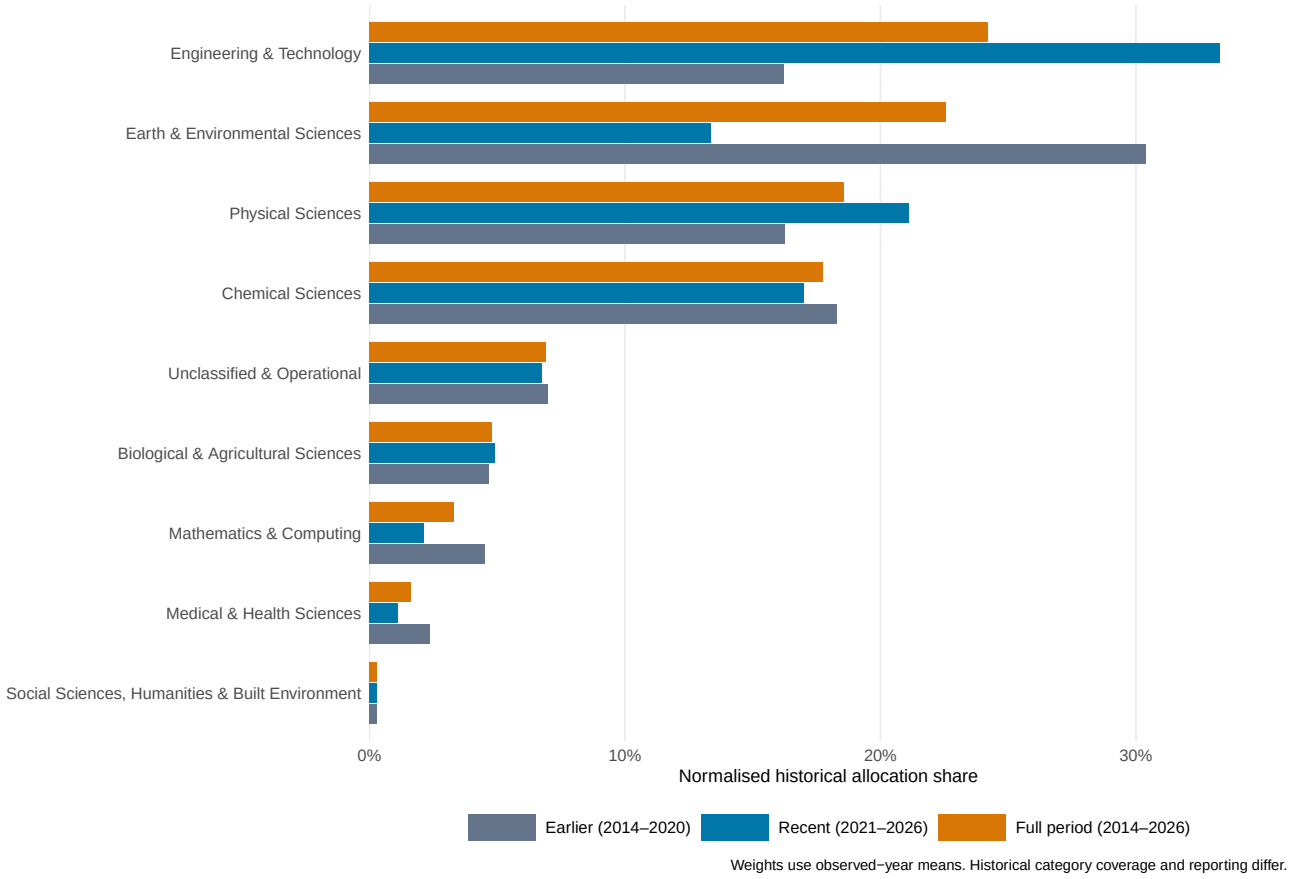


Figure 14: Normalised Field of Research allocation shares across three historical reference periods.

Table 10 reports the normalised scenario weights for all nine research-field groups. The number of reported years is also included to show where the historical coverage differs.

Table 10: Normalised FoR allocation weights and reporting coverage across the three historical reference periods.

Research field	Earlier (2014–2020)	Recent (2021–2026)	Full period (2014–2026)
Biological & Agricultural Sciences	4.69% (7)	4.93% (6)	4.81% (13)
Chemical Sciences	18.28% (7)	17.02% (6)	17.74% (13)
Earth & Environmental Sciences	30.38% (7)	13.36% (6)	22.55% (13)
Engineering & Technology	16.24% (7)	33.28% (6)	24.19% (13)
Mathematics & Computing	4.51% (6)	2.13% (6)	3.32% (12)
Medical & Health Sciences	2.38% (4)	1.12% (6)	1.63% (10)
Physical Sciences	16.26% (7)	21.11% (6)	18.55% (13)
Social Sciences, Humanities & Built Environment	0.30% (1)	0.30% (4)	0.30% (5)
Unclassified & Operational	6.98% (5)	6.75% (2)	6.92% (7)

Note: Each cell shows the normalised allocation weight followed by the number of years with reported observations in parentheses. The reference periods contain seven, six and thirteen years respectively. Percentages are normalised within each period.

The earlier-period scenario gives the largest allocation weight to Earth and Environmental Sciences at approximately 30.38%. Chemical Sciences follows at 18.28%, with Physical Sciences and Engineering and Technology accounting for 16.26% and 16.24% respectively.

The recent-period composition differs considerably. Engineering and Technology becomes the largest group at 33.28%, followed by Physical Sciences (21.11%), Chemical Sciences (17.02%) and Earth and Environmental Sciences (13.36%).

Engineering and Technology therefore gains approximately 17.04 percentage points between the earlier and recent scenarios. Over the same comparison, Earth and Environmental Sciences loses approximately 17.02 percentage points.

The full-period weights lie between these two compositions for the major research fields. Engineering and Technology accounts for approximately 24.19%, Earth and Environmental Sciences for 22.55%, Physical Sciences for 18.55% and Chemical Sciences for 17.74%.

Reporting coverage also differs across research fields. For example, the recent-period weight for Unclassified and Operational is based on only two reported years, while most major research fields have observations for all six years. Social Sciences, Humanities and Built Environment has only five reported years across the full period. Because the weights are calculated from available observations, groups with incomplete coverage may not represent the entire reference period.

These differences should not automatically be interpreted as changes in underlying research demand. The historical records span the FoR 2008 and FoR 2020 classification systems. In addition, the 2017

Pawsey-wide reporting scope differs from other observations, while the completeness of category reporting varies across years.

The normalised weights therefore provide a consistent calculation method for comparing the constructed scenarios, but they do not remove differences in classification, facility coverage or reporting completeness. Nor do they establish that research priorities or computing requirements changed by the same percentages.

4.4.3 Illustrative Distribution of Demand in 2030

The three historical allocation compositions were applied to the same selected national HPC demand projection.

The central linear model projects approximately 2.50 million kSU in 2030. Each scenario distributes this national total across the nine research-field groups using its corresponding normalised weights.

Table 11 presents the resulting distributions. Calculations use the unrounded national forecast.

Table 11: Illustrative 2030 HPC demand distribution under three historical FoR allocation scenarios, in kSU.

Research field	Earlier (2014–2020)	Recent (2021–2026)	Full period (2014–2026)
Engineering & Technology	406,401	832,908	605,405
Physical Sciences	406,985	528,466	464,339
Chemical Sciences	457,455	425,998	443,903
Earth & Environmental Sciences	760,374	334,353	564,291
Unclassified & Operational	174,599	168,853	173,130
Biological & Agricultural Sciences	117,258	123,418	120,389
Mathematics & Computing	112,832	53,320	83,184
Medical & Health Sciences	59,462	28,081	40,714
Social Sciences, Humanities & Built Environment	7,455	7,426	7,467

The choice of historical reference period produces substantial differences in the indicative distribution of national HPC demand. The national forecast remains unchanged between scenarios; only the assumed distribution differs.

Under the recent-period scenario, Engineering and Technology accounts for approximately 832,908 kSU in 2030, compared with 406,401 kSU under the earlier-period scenario. This is a difference of 426,507 kSU, even though both scenarios use the same national demand projection.

Earth and Environmental Sciences shows the opposite pattern. Its indicative allocation decreases from 760,374 kSU under the earlier-period scenario to 334,353 kSU under the recent-period scenario, a difference of 426,021 kSU.

Under the recent-period scenario, Physical Sciences accounts for approximately 528,466 kSU, while Chemical Sciences accounts for 425,998 kSU. The corresponding full-period estimates are 464,339 kSU and 443,903 kSU, respectively.

These differences arise entirely from the historical allocation weights applied to the same national forecast. The choice of reference period is therefore an important modelling assumption, not merely a presentation choice.

The figures should not be interpreted as confirmed future allocations or direct forecasts of researcher demand. They are conditional estimates showing how the national forecast would be distributed if each historical allocation composition were maintained.

4.4.4 Interpretation and Planning Implications

The three scenarios provide different views of how projected national HPC demand might be distributed.

The earlier-period scenario reflects a composition in which Earth and Environmental Sciences received a larger share of allocations. The recent-period scenario places greater weight on Engineering and Technology. The full-period scenario provides a broader historical reference, although it remains affected by reporting differences across the complete series.

The results also demonstrate an important limitation of using allocation records as a proxy for demand. Allocations reflect computing resources made available through allocation processes, which may be lower than the resources requested or required by researchers. Therefore, the indicative distributions should not be interpreted as direct measures of future research-field demand.

The FoR classification change and differences in facility reporting also limit how confidently historical shifts can be attributed to changes in research activity.

For national HPC planning, the main implication is that the assumed research-field composition can materially change the indicative distribution of a fixed national forecast. The difference of approximately 426,500 kSU for Engineering and Technology illustrates this sensitivity. Comparing alternative reference periods makes the uncertainty more visible than selecting one historical composition as definitive.

More consistent FoR-level records of requested, allocated and used computing resources would support stronger future analysis. A validated mapping between historical classification systems and more complete reporting across facilities would also improve the reliability of research-field comparisons.

Overall, the FoR scenarios extend the selected national HPC forecast by showing how different historical allocation assumptions affect its illustrative distribution. They provide a basis for examining planning sensitivity, but they do not establish independently validated forecasts of future demand for individual research fields.

4.5 Summary of Results

The historical analysis identified increasing pressure between recorded HPC demand and available allocations. In 2026, recorded HPC demand reached approximately 2.24 million kSU, while allocations were approximately 872,000 kSU. This represented a fulfilment rate of 38.9%, indicating that recorded requests substantially exceeded allocated resources.

The rolling-origin evaluation identified the linear trend as the preferred central model as the central forecasting model for overall HPC and CPU demand. By 2030, these models project approximately 2.50 million kSU and 1.37 billion CPU core-hours, respectively. Alternative growth assumptions produce

substantially different projections, particularly under the log-linear models. The results are also sensitive to the inclusion of the 2026 observation.

GPU forecasting remained exploratory because only five annual modelling observations were available. The five scenarios produced 2030 estimates ranging from approximately 2.25 to 3.68 million GPU-hours. No central GPU forecast was selected, and these projections were excluded from the initial national HPC planning scenarios.

The FoR scenarios demonstrated that the indicative distribution of projected national demand depends strongly on historical allocation assumptions. Engineering and Technology accounted for 16.24% of the earlier-period scenario compared with 33.28% of the recent-period scenario. Applying these weights to the same national forecast produced a difference of approximately 426,500 kSU in its illustrative 2030 demand distribution.

Together, these findings show that forecasting results depend not only on historical model accuracy but also on data coverage, measurement assumptions, growth specifications and scenario construction. Their implications for national HPC planning are discussed in the following chapter.

5 Discussion

5.1 Interpretation of the Main Findings

The analysis identified three main findings: recorded HPC demand has grown faster than available allocations, longer-term forecasts are sensitive to modelling assumptions, and the distribution of projected demand across research fields depends on the historical allocation period used.

5.1.1 Growing Difference Between Demand and Allocation

The historical analysis showed a substantial difference between requested and allocated HPC resources. In 2026, recorded demand reached approximately 2.24 million kSU, while allocations were approximately 872,000 kSU. This resulted in a fulfilment rate of 38.9%, indicating that less than half of the recorded requests were allocated.

This difference is important because increasing demand does not necessarily result in a corresponding increase in available computing resources. If requests continue growing while allocations remain constrained, the gap between the two measures could become more pronounced.

However, the recorded gap should not be interpreted as a complete measure of unmet national research demand. Researchers may adjust their applications according to expected availability, while others may choose not to apply because of capacity limitations. The available records also do not show whether all requested resources would have been used if allocated.

Therefore, the historical results provide evidence of pressure within the reported allocation system, rather than a direct estimate of the total computing capacity required by Australian researchers.

5.1.2 Interpretation of the Forecasting Results

The rolling-origin evaluation identified the linear trend as the preferred central model for overall HPC and CPU demand. These models provided competitive historical accuracy without assuming that

proportional growth would continue indefinitely.

The selected models project approximately 2.50 million kSU of overall HPC demand and 1.37 billion CPU core-hours by 2030. Both projections indicate continued growth, although their interpretation depends on the historical measures used to construct the datasets.

The alternative scenarios demonstrate why the central forecasts should not be treated as the only possible outcomes. For overall HPC demand, excluding the 2026 observation reduces the 2030 projection from approximately 2.50 million to 2.29 million kSU. The full-history log-linear model produces a substantially higher projection of approximately 6.48 million kSU.

CPU demand shows a similar sensitivity. The linear model projects approximately 1.37 billion core-hours in 2030, compared with approximately 2.48 billion under the full-history log-linear model.

An important difference is that the log-linear model performed poorly relative to the linear model for overall HPC demand, while the two approaches performed similarly for CPU demand. Consequently, the higher CPU scenario cannot be dismissed solely on the basis of historical forecast accuracy. Nevertheless, competitive one-year-ahead performance does not establish that proportional growth will continue through 2030.

These findings show that model selection involves more than choosing the lowest historical forecast error. The assumed growth pattern becomes increasingly important as the forecasting horizon extends.

The statistical prediction intervals provide another perspective on uncertainty. For 2030, the selected overall HPC model has a 95% prediction interval of approximately 1.98–3.03 million kSU, while the corresponding CPU interval is approximately 974–1,766 million core-hours. These intervals are conditional on the fitted linear models and do not include uncertainty arising from alternative growth specifications, historical conversions or future technological changes.

5.1.3 GPU Demand and Research Field Differences

GPU forecasting presented a different challenge. The modelling dataset contained only five annual observations from 2022 to 2026, and the rolling-origin evaluation covered just two years. Although the naïve benchmark achieved the lowest forecast errors, this provides insufficient evidence that GPU demand will remain constant.

The five exploratory scenarios produced 2030 projections ranging from approximately 2.25 to 3.68 million GPU-hours. The substantial difference between models fitted with and without 2026 demonstrates the influence of a single recent observation on such a short series.

For this reason, no central GPU forecast was selected. The scenarios are useful for understanding sensitivity to modelling assumptions, but they are not sufficiently reliable to support specific GPU capacity requirements.

The Field of Research (FoR) analysis identified a different source of uncertainty. Engineering and Technology accounted for 16.24% of the earlier-period allocation scenario but 33.28% of the recent-period scenario. When these weights were applied to the same 2030 national HPC demand projection, the indicative distribution for this field differed by approximately 426,500 kSU.

Earth and Environmental Sciences showed the opposite pattern, receiving a considerably greater share under the earlier-period scenario.

These differences demonstrate that the choice of historical reference period can substantially affect the indicative distribution of projected national demand. However, they cannot establish that underlying research demand changed by the same proportions, particularly given the changes in FoR classification and historical reporting coverage.

5.2 Implications for National HPC Planning

5.2.1 Planning for Growth and Uncertainty

The central forecasts provide a quantitative reference for considering future national HPC requirements. However, planning should not depend on a single projected value for 2030.

For overall HPC demand, the selected linear model projects approximately 2.50 million kSU, while alternative assumptions produce substantially different results. This suggests that infrastructure planning should examine a range of possible demand outcomes rather than assuming that the central trajectory will occur exactly as estimated.

The distinction between prediction intervals and alternative growth scenarios is particularly important. Prediction intervals describe uncertainty conditional on a selected statistical model, whereas alternative scenarios demonstrate how different historical samples or growth assumptions change the projected demand.

Both forms of uncertainty are relevant, but neither captures every factor affecting future infrastructure requirements.

The historical difference between requested and allocated resources also needs to be considered. Projected demand alone does not show how much additional capacity is necessary because the available records do not establish a direct relationship between requested kSU, allocated resources and actual usage.

Consequently, the forecasts are most suitable as inputs to planning discussions and further capacity assessments, rather than direct specifications for infrastructure investment.

5.2.2 Accounting for Changes in Computing Technology

Another important consideration is that HPC demand is reported using different accounting measures.

Overall HPC demand is expressed in kSU, while CPU and GPU demand are represented using core-hours and GPU-hours. These measures describe different aspects of computing activity and should not be added together or treated as directly interchangeable.

Changes in hardware performance, system configurations and service-unit conversion factors can also affect the relationship between reported demand and physical computing capacity.

For example, a future computing system may deliver greater performance per core-hour than an earlier system. Similarly, changes in GPU architectures could affect the resources needed to complete comparable research workloads.

This means that an increase in reported computing demand does not necessarily translate into an identical percentage increase in physical infrastructure requirements.

National planning would therefore benefit from combining demand projections with information about system performance, resource utilisation, workload characteristics and expected technological changes.

5.2.3 Considering Research Field Requirements

The FoR scenarios provide an additional perspective by illustrating how the same national HPC projection can be distributed differently across research disciplines.

The recent-period scenario gives greater weight to Engineering and Technology, while the earlier-period scenario gives greater weight to Earth and Environmental Sciences. These differences may be useful when examining how historical allocation patterns influence indicative future resource distributions.

However, allocation shares are not direct measures of research demand. They reflect resources made available through historical allocation processes and may be affected by capacity constraints, reporting practices and classification changes.

For this reason, the FoR scenarios should not be used as fixed allocation targets or as evidence that particular disciplines will require the projected amounts.

Instead, they provide a basis for discussing distributional sensitivity and identifying where more detailed research-field demand information would improve planning.

5.3 Limitations and Future Work

5.3.1 Data Availability and Historical Comparability

The main limitation of the analysis is the short and uneven historical record.

Overall HPC and CPU demand were modelled using 13 annual observations from 2014 to 2026. Although sufficient to compare simple forecasting methods, this provides limited information for estimating longer-term growth patterns or identifying structural changes.

GPU forecasting was more restricted, with only five annual modelling observations. The limited evaluation period prevented a reliable comparison of long-term forecasting performance.

Historical comparability also presented challenges. The source records covered different facilities, computing systems and reporting conventions. Resource conversions were required to construct consistent analytical measures, and some GPU estimates depended on allocation shares and working assumptions.

Although these assumptions were documented, they introduce measurement uncertainty that is not fully represented in the statistical forecast intervals.

The FoR analysis was also affected by differences between the 2008 and 2020 classification systems, changes in reporting coverage and the inclusion of records with different facility scopes.

Normalisation allowed historical allocation shares to be compared across the constructed scenarios, but it did not remove these underlying differences.

5.3.2 Forecasting and Scenario Limitations

The forecasting methods were intentionally simple because of the limited annual observations.

Linear and log-linear models provide interpretable representations of historical growth, but they cannot directly account for future changes in research funding, computing technology, national policy or facility capacity.

The rolling-origin evaluation assessed one-year-ahead forecasting performance. Strong performance over this horizon does not necessarily imply reliable projections four years into the future.

The results were also sensitive to the 2026 observation. This was particularly evident in the alternative HPC, CPU and GPU scenarios, demonstrating that a single recent year can materially influence longer-term projections.

Furthermore, the model-based prediction intervals depend on the assumptions of the selected linear models. They do not account for uncertainty in historical reconstruction, future technology or the choice between alternative forecasting specifications.

The FoR scenarios have a separate limitation because they distribute national demand using historical allocation shares rather than independently forecasting demand within each research field.

5.3.3 Recommendations for Future Work

Future analysis could be strengthened through several improvements:

1. **Improve requested, allocated and used resource records.** Consistent annual data across facilities would make it possible to examine the relationship between applications, allocations and actual utilisation more directly.
2. **Validate resource conversion factors.** Further consultation with facility operators could improve the consistency of historical kSU, CPU and GPU measures, particularly where hardware configurations or accounting rules changed.
3. **Extend the GPU demand series.** More detailed GPU request, allocation and usage records would support stronger evaluation and help determine whether a separate GPU forecasting model is appropriate.
4. **Improve FoR-level demand information.** Consistent research-field request data and validated classification mappings would allow demand to be modelled directly rather than approximated using historical allocation shares.
5. **Update and evaluate forecasts regularly.** As new annual observations become available, the models should be re-estimated and their forecast performance reassessed. This would also help determine whether the linear trends remain appropriate.
6. **Connect demand projections with infrastructure characteristics.** Combining forecasting results with hardware performance, utilisation and expected technology changes would provide a stronger basis for translating reported computing demand into practical capacity requirements.

Overall, the project provides a documented and reproducible basis for examining historical national HPC demand and its possible development through 2030. Its main contribution is not a single definitive capacity estimate, but a comparison of plausible projections, their underlying assumptions and the limitations of the available evidence.

These findings can support national HPC planning discussions while identifying the additional data required for more reliable future assessments.

6 Conclusion

This project assembled and standardised publicly available Australian national research HPC records covering 2014–2026 and developed demand projections through 2030. The analysis combined historical data reconstruction, resource conversion, descriptive analysis, rolling-origin forecast evaluation and scenario development. It also documented the assumptions and limitations associated with using publicly reported computing data.

The historical analysis identified a substantial difference between requested and allocated HPC resources. In 2026, recorded demand reached approximately 2.24 million kSU, compared with allocations of approximately 872,000 kSU, representing a fulfilment rate of 38.9%. This finding indicates increasing pressure within the reported allocation system, although the difference cannot be treated as a complete measure of unmet national research demand.

The forecasting evaluation selected linear trend models as the central approach for overall HPC and CPU demand. By 2030, these models project approximately 2.50 million kSU of overall HPC demand and 1.37 billion CPU core-hours. Alternative models and the exclusion of the 2026 observation produced substantially different projections, demonstrating the importance of historical data coverage and growth assumptions when interpreting longer-term forecasts.

GPU projections remained exploratory because only five annual modelling observations were available. The Field of Research scenarios also demonstrated that the indicative distribution of future national HPC demand depends on the historical allocation period selected. These findings highlight the need to distinguish between modelled demand, historical allocations and actual computing capacity.

The main contribution of this project is a reproducible analytical framework that brings together historical data, documented conversion assumptions, forecast evaluation and scenario analysis. The results provide the Australian Academy of Science with an evidence-based reference for discussing future national HPC requirements, while making the uncertainty and data limitations clear. The accompanying Quarto documentation and Git repository preserve the detailed methods and calculations for future review and updates.

7 References

Australian Academy of Science. 2024. "The Future Computing Needs of the Australian Science Sector." https://www.science.org.au/.

Australian Academy of Science. 2026. *Internship Project Brief: Forecasting Demand for National Research Supercomputing.*

Australian Bureau of Statistics. 2020. "Australian and New Zealand Standard Research Classification (ANZSRC), 2020." https://www.abs.gov.au/statistics/classifications/australian-and-new-zealand-standard-research-classification-anzsrc/2020.

Hyndman, Rob J., and George Athanasopoulos. 2021. *Forecasting: Principles and Practice.* 3rd ed. OTexts. https://otexts.com/fpp3/.

Hyndman, Rob J., and Anne B. Koehler. 2006. "Another Look at Measures of Forecast Accuracy." *International Journal of Forecasting* 22 (4): 679–88. https://doi.org/10.1016/j.ijforecast.2006.03.001.

National Computational Infrastructure. 2014a. *Annual Report 2013/14*. https://nci.org.au/sites/default/files/annual-reports/2019-07/About%20Us_Annual%20Reports_2013-14.pdf.

National Computational Infrastructure. 2014b. "NCMAS Information for Applicants." https://nci.org.au/news-events/news/ncmas-information-applicants.

National Computational Infrastructure. 2015. *Annual Report 2014/15*. https://nci.org.au/sites/default/files/annual-reports/2019-07/About%20Us_Annual%20Reports_2014-15.pdf.

National Computational Infrastructure. 2016. *Annual Report 2015/16*. https://nci.org.au/sites/default/files/annual-reports/2019-07/About%20Us_Annual%20Reports_2015-16.pdf.

National Computational Infrastructure. 2017. *Annual Report 2016/17*. https://nci.org.au/sites/default/files/annual-reports/2019-07/About%20Us_Annual%20Reports_2016-17.pdf.

National Computational Infrastructure. 2018. *Annual Report 2017/18*. https://nci.org.au/sites/default/files/annual-reports/2019-07/About%20Us_Annual%20Reports_2017-18.pdf.

National Computational Infrastructure. 2019. *Annual Report 2018/19*. https://nci.org.au/sites/default/files/annual-reports/2019-11/NCI%20Report%202019%20web.pdf.

National Computational Infrastructure. 2020a. *Annual Report 2019/20*. https://nci.org.au/sites/default/files/annual-reports/2021-01/NCI%20Annual%20report%202019-20%20web_final_opt2.pdf.

National Computational Infrastructure. 2020b. *NCMAS 2020 Information for Applicants*. https://nci.org.au/sites/default/files/documents/2019-10/NCMAS-2020-InformationForApplicants.pdf.

National Computational Infrastructure. 2022. *NCMAS 2023 Information for Applicants*. https://my.nci.org.au/mancini-assets/media/mas/ncmas/2023/NCMAS2023-NCMAS_2023_Info4Applicants_v5-20220830T164026AEST.pdf.

National Computational Infrastructure. 2023. *NCMAS 2024 Information for Applicants*. https://my.nci.org.au/mancini-assets/media/mas/ncmas/2024/NCMAS2024-NCMAS_2024_-_Information_for_Applicants_20231109-20231109T131740AEDT.pdf.

National Computational Infrastructure. 2025. *NCMAS 2026 Information for Applicants*. https://my.nci.org.au/mancini-assets/media/mas/ncmas/2026/NCMAS2026-Information_for_Applicants-20251023T122535AEDT.pdf.

National Computational Infrastructure. n.d.-a. *HPC Systems*. Accessed October 18, 2026. https://nci.org.au/infrastructure/hpc-systems.

National Computational Infrastructure. n.d.-b. "Who We Are." Accessed October 4, 2026. https://nci.org.au/.

National Computational Merit Allocation Scheme. 2021. "NCMAS 2021 Outcomes." https://my.nci.org.au/mancini/ncmas/2021/outcomes.

National Computational Merit Allocation Scheme. 2022. "NCMAS 2022 Outcomes." https://my.nci.org.au/mancini/ncmas/2022/outcomes.

National Computational Merit Allocation Scheme. 2023. "NCMAS 2023 Outcomes." https://my.nci.org.au/mancini/ncmas/2023/outcomes.

National Computational Merit Allocation Scheme. 2024. "NCMAS 2024 Outcomes." https://my.nci.org.au/mancini/ncmas/2024/outcomes.

National Computational Merit Allocation Scheme. 2025. "NCMAS 2025 Outcomes." https://my.nci.org.au/mancini/ncmas/2025/outcomes.

National Computational Merit Allocation Scheme. 2026. "NCMAS 2026 Outcomes." https://my.nci.org.au/mancini/ncmas/2026/outcomes.

Pawsey Supercomputing Research Centre. 2017. *Annual Report 2016/17*. https://pawsey.org.au/wp-content/uploads/2018/10/Pawsey-Annual-Report-2016-2017.pdf.

Pawsey Supercomputing Research Centre. n.d.-a. "About Pawsey." Accessed October 4, 2026. https://pawsey.org.au/.

Pawsey Supercomputing Research Centre. n.d.-b. *Changes in Supercomputing Services for 2023*. Accessed October 18, 2026. https://pawsey.atlassian.net/wiki/spaces/US/pages/51927586/Changes+in+Supercomputing+Services+for+2023.

Pawsey Supercomputing Research Centre. n.d.-c. *Setonix Fires up for Researchers*. Accessed October 18, 2026. https://pawsey.org.au/setonix-fires-up-for-researchers/.

Tashman, Leonard J. 2000. "Out-of-Sample Tests of Forecasting Accuracy: An Analysis and Review." *International Journal of Forecasting* 16 (4): 437–50. https://doi.org/10.1016/S0169-2070(00)00065-0.